Perturbations in Celestial Mechanics

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November 10, 2026

Preface

This document contains the result of my frustration associated to the presentation of perturbation theory in standard textbooks on celestial mechanics [1], [2]. In these books, the intermediate steps between equations are typically far from obvious and require (very) lenghty derivations. Consequently, this document expands on the intermediate steps so that one should be able to follow the computation fully with mental arithmetic. I'm not an expert on the topic so it is very likely that the document will contain some conceptual errors. Moreover, no AI or any other system has been used for the calculations or verification of their validity.

*https://www.github.com/vsr83

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1 Planetary Disturbing Function

1.1 Two-Planet Solar System

Consider a solar system with two planets orbiting a star. The position vectors and masses of the two planets and the star are denoted with $\boldsymbol{R}$, $\boldsymbol{R}'$, $\boldsymbol{R}_s$ and m, m', M, respectively. Then, from Newton's laws of motion, we obtain

$$\ddot{\boldsymbol{R}}_s = Gm\frac{\boldsymbol{R}-\boldsymbol{R}_s}{|\boldsymbol{R}-\boldsymbol{R}_s|^3} + Gm'\frac{\boldsymbol{R}'-\boldsymbol{R}_s}{|\boldsymbol{R}'-\boldsymbol{R}_s|^3} \tag{1}$$

$$\ddot{\boldsymbol{R}} = Gm'\frac{\boldsymbol{R}'-\boldsymbol{R}}{|\boldsymbol{R}'-\boldsymbol{R}|^3} + GM\frac{\boldsymbol{R}_s-\boldsymbol{R}}{|\boldsymbol{R}_s-\boldsymbol{R}|^3} \tag{2}$$

$$\ddot{\boldsymbol{R}}' = Gm\frac{\boldsymbol{R}-\boldsymbol{R}'}{|\boldsymbol{R}-\boldsymbol{R}'|^3} + GM\frac{\boldsymbol{R}_s-\boldsymbol{R}'}{|\boldsymbol{R}_s-\boldsymbol{R}'|^3}, \tag{3}$$

where G is the gravitational constant. Definition of position vectors $\boldsymbol{r} := \boldsymbol{R} - \boldsymbol{R}_s$ and $\boldsymbol{r}' := \boldsymbol{R}' - \boldsymbol{R}_s$ relative to the position vector of the star, yields

$$\ddot{\boldsymbol{r}} = \ddot{\boldsymbol{R}} - \ddot{\boldsymbol{R}}_s \tag{4}$$

$$= Gm'\frac{\boldsymbol{R}'-\boldsymbol{R}}{|\boldsymbol{R}'-\boldsymbol{R}|^3} + GM\frac{\boldsymbol{R}_s-\boldsymbol{R}}{|\boldsymbol{R}_s-\boldsymbol{R}|^3} - Gm\frac{\boldsymbol{R}-\boldsymbol{R}_s}{|\boldsymbol{R}-\boldsymbol{R}_s|^3} - Gm'\frac{\boldsymbol{R}'-\boldsymbol{R}_s}{|\boldsymbol{R}'-\boldsymbol{R}_s|^3} \tag{5}$$

$$= Gm'\left(\frac{\boldsymbol{r}'-\boldsymbol{r}}{|\boldsymbol{r}'-\boldsymbol{r}|^3} - \frac{\boldsymbol{r}'}{r'^3}\right) - \mu\frac{\boldsymbol{r}}{r^3}, \tag{6}$$

where we have defined the **standard gravitational parameter** $\mu := G(M+m)$. Reorganizing and performing the same derivation for $\ddot{\boldsymbol{r}}'$, we obtain the equations for the **three-body problem**

$$\ddot{\boldsymbol{r}} + \mu\frac{\boldsymbol{r}}{r^3} = Gm'\left(\frac{\boldsymbol{r}'-\boldsymbol{r}}{|\boldsymbol{r}'-\boldsymbol{r}|^3} - \frac{\boldsymbol{r}'}{r'^3}\right) \tag{7}$$

$$\ddot{\boldsymbol{r}}' + \mu'\frac{\boldsymbol{r}'}{r'^3} = Gm\left(\frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|^3} - \frac{\boldsymbol{r}}{r^3}\right), \tag{8}$$

where the left-hand side represents a two-body problem and the right-hand a **perturbation** from a third body. When the mass M is significantly greater than the masses m and m' and the distances between the two smaller masses are large, the perturbation stays small.

1.1.1 Disturbing Functions

We represent the perturbations as the gradients $\nabla\mathcal{R}$ and $\nabla\mathcal{R}'$ of the **disturbing functions** $\mathcal{R}$ and $\mathcal{R}'$. That is,

$$\nabla\mathcal{R} = Gm'\left(\frac{\boldsymbol{r}'-\boldsymbol{r}}{|\boldsymbol{r}'-\boldsymbol{r}|^3} - \frac{\boldsymbol{r}'}{r'^3}\right), \tag{9}$$

$$\nabla'\mathcal{R}' = Gm\left(\frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|^3} - \frac{\boldsymbol{r}}{r^3}\right), \tag{10}$$

where ∇' represents gradient w.r.t. primed coordinates $(\partial/\partial x', \partial/\partial y', \partial/\partial z')$. To obtain an expression for the disturbing functions, note that

$$\frac{\partial}{\partial x}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} = \frac{\partial}{\partial x}\frac{1}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}} = -\frac{x-x'}{|\boldsymbol{r}-\boldsymbol{r}'|^3}. \tag{11}$$

Thus, obviously also

$$\nabla\left(Gm'\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|}\right) = -Gm'\frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|^3} = Gm'\frac{\boldsymbol{r}'-\boldsymbol{r}}{|\boldsymbol{r}'-\boldsymbol{r}|^3}, \tag{12}$$

$$\nabla'\left(Gm\frac{1}{|\boldsymbol{r}'-\boldsymbol{r}|}\right) = -Gm\frac{\boldsymbol{r}'-\boldsymbol{r}}{|\boldsymbol{r}'-\boldsymbol{r}|^3} = Gm\frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|^3}. \tag{13}$$

Similarly, we can note that

$$\frac{\partial}{\partial x}\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r'^3} = \frac{x'}{r'^3} \quad \Rightarrow \quad \nabla\left(\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r'^3}\right) = \frac{\boldsymbol{r}'}{r'^3}. \tag{14}$$

to obtain expressions for the disturbing function

$$\mathcal{R} = \overline{\mu}'\left(\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} - \frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r'^3}\right) \tag{15}$$

$$\mathcal{R}' = \overline{\mu}\left(\frac{1}{|\boldsymbol{r}'-\boldsymbol{r}|} - \frac{\boldsymbol{r}'\cdot\boldsymbol{r}'}{r^3}\right), \tag{16}$$

where we have defined $\overline{\mu} := Gm$ and $\overline{\mu}' := Gm'$.

The disturbing functions will be now transformed to expressions involving only the angle between the location vectors and the ratios between the semi-major axis and distance from the central body. This will allow easy expansion in terms of orbital elements. Let us denote

$$\mathcal{S} := \frac{a'}{\overline{\mu}'}\mathcal{R} = \mathcal{S}_D + \alpha\mathcal{S}_E \tag{17}$$

$$\mathcal{S}' := \frac{a}{\overline{\mu}}\mathcal{R}' = \alpha\mathcal{S}_D + \alpha^{-1}\mathcal{S}_I, \tag{18}$$

where $\alpha := a/a'$ and

$$\mathcal{S}_D = \frac{a'}{|\boldsymbol{r}-\boldsymbol{r}'|} = a'\left(r'^2 - 2r'r\cos\psi + r^2\right)^{-1/2} \tag{19}$$

$$\mathcal{S}_E = -\frac{a'}{\overline{\mu}'}\left(\frac{a'}{a}\right)\frac{rr'\cos\psi}{r'^2} = -\left(\frac{r}{a}\right)\left(\frac{a'}{r'}\right)^2\cos\psi, \tag{20}$$

$$\mathcal{S}_I = -\frac{a}{\overline{\mu}}\left(\frac{a}{a'}\right)\frac{r'r\cos\psi}{r^2} = -\left(\frac{r'}{a'}\right)\left(\frac{a}{r}\right)^2\cos\psi \tag{21}$$

where ψ is the angle between the vectors $\boldsymbol{r}$ and $\boldsymbol{r}'$.

1.2 Expansion of S_D

Let us denote

$$\zeta := \frac{r-a}{a}, \quad \zeta' := \frac{r'-a'}{a'}, \quad \delta := \cos\psi - \cos(\theta-\theta'), \tag{22}$$

where $\theta = \varpi + f, \theta' = \varpi' + f'$ are the true longitudes for the two planets.

The variables ζ and ζ' essentially measure how many multiples of semi-major axis the corresponding body is away from a circular orbit with with semi-major axis a. For nearly circular orbits, ζ, ζ' are very small. In general, for elliptic orbits $r_{max} = a(1+e)$ and $\zeta_{max} = [a(1+e)-a]/a = e$. The variable δ disappears when the planets orbit on the same plane and measures the difference in ψ due to difference in orbital planes. We assume that the inclinations and eccentrities of both planets are small, which suggests that the effect of e and I can be included using a series expansion.

The equations (19)-(21) are first expanded in terms of Keplerian elements $e, \lambda, \Omega, \varpi$ and $s := \sin(I/2)$. The series expansions here are done up to the second order of eccentricity or inclination. That is, we only include terms proportional to $e^{k_1}e'^{k_2}s^{k_3}s'^{k_4}$, where $k_1 + k_2 + k_3 + k_4 \le 2$. To avoid too complicated expressions in equations, we often compute parts of an expression involving specific orders of (e, e, s, s') separately and use a dedicated notation to indicate this. For example, $\cos[j(\theta-\theta')]$ will involve orders $1, e, e', ee', e^2, e'^2$. The part of the expression that is proportional to e^2, will be denoted by $\cos[j(\theta-\theta')]_{e^2}$. That is, in a second order expression

$$\begin{aligned}\cos[j(\theta-\theta')] &= \cos[j(\theta-\theta')]_1 \\ &+ e\cos[j(\theta-\theta')]_e + e'\cos[j(\theta-\theta')]_{e'} \\ &+ ee'\cos[j(\theta-\theta')]_{ee'} + e^2\cos[j(\theta-\theta')]_{e^2} + e'^2\cos[j(\theta-\theta')]_{e'^2}.\end{aligned} \tag{23}$$

1.2.1 Taylor Series Expansion of S_D

Substitution of $a' = r'/(1+\zeta')$ to (19), yields

$$S_D = \frac{1}{1+\zeta'}\left(\frac{r^2}{r'^2} - 2\frac{r}{r'}\cos\psi + 1\right)^{-1/2}. \tag{24}$$

The first two terms can be expanded

$$\frac{r}{r'} = \frac{a(1+\zeta)}{a'(1+\zeta')} = \alpha\frac{1+\zeta}{1+\zeta'}, \tag{25}$$

$$2\frac{r}{r'}\cos\psi = 2\alpha\frac{1+\zeta}{1+\zeta'}\cos\psi = 2\alpha\frac{1+\zeta}{1+\zeta'}\delta - 2\alpha\frac{1+\zeta}{1+\zeta'}\cos(\theta-\theta') \tag{26}$$

to obtain

$$S_D = \left[\alpha^2\left(\frac{1+\zeta}{1+\zeta'}\right)^2 - 2\alpha\frac{1+\zeta}{1+\zeta'}\delta - 2\alpha\frac{1+\zeta}{1+\zeta'}\cos(\theta-\theta') + 1\right]^{-1/2}. \tag{27}$$

When the orbits are circular, the expressions $(1+\zeta)/(1+\zeta')$ simplify to unity. Since the planetary orbits are nearly circular and are close to the same plane, it makes expand (27) as a Taylor series w.r.t. perfectly circular orbits on the zero-inclination plane. Define

$$\tilde{\alpha} := \frac{1+\zeta}{1+\zeta'}\alpha. \tag{28}$$

Define

$$F(\alpha, \Delta, \theta - \theta') := \left[1 + \alpha^2 - 2\alpha\Delta - 2\alpha\cos(\theta - \theta')\right]^{-1/2}. \tag{29}$$

Then, (27) can be rewritten $S_D = F(\tilde{\alpha}, \delta, \theta - \theta')$ and circular orbits correspond to $F(\alpha, \delta, \theta - \theta')$ and planets on same zero-inclination plane correspond to $F(\tilde{\alpha}, 0)$. We expand to the second order

$$\begin{aligned} F(\tilde{\alpha}, \delta, \theta - \theta') &\approx F(\alpha, 0, \theta - \theta') + \frac{\partial F}{\partial \alpha}(\alpha, 0, \theta - \theta')(\tilde{\alpha} - \alpha) + \frac{1}{2}\frac{\partial^2 F}{\partial \alpha^2}(\alpha, 0, \theta - \theta')(\tilde{\alpha} - \alpha)^2 \\ &+ \frac{\partial F}{\partial \Delta}(\alpha, 0, \theta - \theta')\delta. \end{aligned} \tag{30}$$

The last term can be evaluated

$$\frac{\partial F}{\partial \Delta}\delta = -\frac{1}{2}\frac{-2\alpha}{[\cdot]^3}\delta = \alpha F^3 \delta \tag{31}$$

Denoting $D := \partial/\partial\alpha$, we can approximate

$$S_D \approx \frac{1}{1+\zeta'}\left[1 + (\tilde{\alpha} - \alpha)D + \frac{1}{2}(\tilde{\alpha} - \alpha)^2 D^2\right] F + \delta\alpha F^3. \tag{32}$$

We also approximate to the second order

$$\frac{1}{1+\zeta'} = \sum_{k=0}^{\infty}(-1)^k \zeta'^k \approx 1 - \zeta' + \zeta'^2 + \mathcal{O}(\zeta'^3), \tag{33}$$

$$\frac{1}{(1+\zeta')^2} = \sum_{k=0}^{\infty}(-1)^k (1+n)\zeta'^k \approx 1 - 2\zeta' + 3\zeta'^2 + \mathcal{O}(\zeta'^3), \tag{34}$$

$$\frac{1}{(1+\zeta')^3} = \frac{1}{2}\sum_{k=0}^{\infty}(-1)^n (n+1)(n+2)\zeta'^n \approx 1 - 3\zeta' + 6\zeta'^2 + \mathcal{O}(\zeta^3) \tag{35}$$

to obtain

$$\frac{\tilde{\alpha} - \alpha}{1+\zeta'} = \frac{\alpha}{1+\zeta'}\left(\frac{1+\zeta}{1+\zeta'} - 1\right) = \alpha\frac{\zeta - \zeta'}{(1+\zeta')^2} \approx \alpha(\zeta - \zeta')(1 - 2\zeta') = \alpha(\zeta - \zeta' + 2\zeta'^2 - 2\zeta\zeta'), \tag{36}$$

$$\frac{(\tilde{\alpha} - \alpha)^2}{1+\zeta'} = \frac{\alpha^2}{1+\zeta'}\left(\frac{1+\zeta}{1+\zeta'} - 1\right)^2 = \alpha^2\frac{(\zeta - \zeta')^2}{(1+\zeta')^3} \approx \alpha^2(\zeta^2 - 2\zeta\zeta' + \zeta'^2)(1 - 3\zeta') \approx \alpha^2(\zeta^2 - 2\zeta\zeta' + \zeta'^2). \tag{37}$$

Therefore, the Taylor series expansion (32) can be written

$$S_D \approx \left[(1 - \zeta' + \zeta'^2) + (\zeta - \zeta' + 2\zeta'^2 - 2\zeta\zeta')\alpha D + \frac{1}{2}(\zeta^2 - 2\zeta\zeta' + \zeta'^2)\alpha^2 D^2\right] F + \delta\alpha F^3 \tag{38}$$

1.2.2 Fourier Series Expansion

Since F is a function of the true longitude difference $\theta - \theta'$, it makes sense to expand the angular dependence as a Fourier series. This dependence is symmetric so the expansion will contain only the cosine

terms. That is,

$$F(\alpha,0,\theta-\theta') = \frac{1}{2}\sum_{j=-\infty}^{\infty} b^{j}_{1/2}(\alpha)\cos[j(\theta-\theta')], \tag{39}$$

$$F^{3}(\alpha,0,\theta-\theta') = \frac{1}{2}\sum_{j=-\infty}^{\infty} b^{j}_{3/2}(\alpha)\cos[j(\theta-\theta')], \tag{40}$$

where **Laplace coefficients**

$$b^{(j)}_{s} := \frac{1}{\pi}\int_{0}^{2\pi}\frac{\cos j\psi\, d\psi}{(1-2\alpha\cos\psi+\alpha^{2})^{s}}. \tag{41}$$

Thereafter, (38) can be expanded

$$\begin{aligned}\mathcal{S}_{D} \approx \frac{1}{2}\sum_{j=-\infty}^{\infty}\Big\{\Big[(1-\zeta'+\zeta'^{2})+(\zeta-\zeta'+2\zeta'^{2}-2\zeta\zeta')\alpha D+\frac{1}{2}(\zeta^{2}-2\zeta\zeta'+\zeta'^{2})\alpha^{2}D^{2}\Big]b^{(j)}_{1/2}(\alpha) \\ + \delta\alpha b^{(j)}_{3/2}(\alpha)\Big\}\cos[j(\theta-\theta')].\end{aligned} \tag{42}$$

We wish to expand the operator in square brackets as a function $G(\lambda,\lambda',\varpi,\varpi',\alpha,\alpha')$. Thus, employ the approximations

$$\zeta = -e\cos(\lambda-\varpi)+\frac{1}{2}e^{2}-\frac{1}{2}e^{2}\cos(2\lambda-2\varpi)+\mathcal{O}(e^{3}), \tag{43}$$

$$\zeta^{2} = e^{2}\cos^{2}(\lambda-\varpi)+\mathcal{O}(e^{3}) \tag{44}$$

$$\approx \frac{1}{2}e^{2}+\frac{1}{2}e^{2}\cos(2\lambda-2\varpi), \tag{45}$$

$$\zeta\zeta' = ee'\cos(\lambda-\varpi)\cos(\lambda'-\varpi')+\mathcal{O}(e^{3}) \tag{46}$$

$$\approx \frac{1}{2}ee'\cos(\lambda-\lambda'-\varpi+\varpi')+\frac{1}{2}ee'\cos(\lambda+\lambda'-\varpi-\varpi') \tag{47}$$

where we have used the trigonometric identity $\cos 2x = 2\cos^{2}x-1$. Thus, we can expand

$$\begin{aligned}1-\zeta'+\zeta'^{2} &= 1+e'\cos(\lambda'-\varpi')-\frac{1}{2}e'^{2}+\frac{1}{2}e'^{2}\cos(2\lambda'-2\varpi') \\ &+ \frac{1}{2}e'^{2}+\frac{1}{2}e'^{2}\cos(2\lambda'-2\varpi') \\ &= 1+e'\cos(\lambda'-\omega')+e'^{2}\cos(2\lambda'-2\omega'),\end{aligned} \tag{48, 49}$$

$$\begin{aligned}\zeta-\zeta'+2\zeta'^{2}-2\zeta\zeta' &= -e\cos(\lambda-\varpi)+\frac{1}{2}e^{2}-\frac{1}{2}e^{2}\cos(2\lambda-2\varpi) \\ &+ e'\cos(\lambda'-\varpi')-\frac{1}{2}e'^{2}+\frac{1}{2}e'^{2}\cos(2\lambda'-2\varpi') \\ &+ e'^{2}+e'^{2}\cos(2\lambda'-2\varpi') \\ &- ee'\cos(\lambda-\lambda'-\varpi+\varpi')-ee'\cos(\lambda+\lambda'-\varpi-\varpi')\end{aligned} \tag{50}$$

$$\begin{aligned}\frac{1}{2}(\zeta^2 - 2\zeta\zeta' + \zeta'^2) &= \frac{1}{4}e^2 + \frac{1}{4}e^2\cos(2\lambda - 2\varpi) \\ &- \frac{1}{2}ee'\cos(\lambda - \lambda' - \varpi + \varpi') - \frac{1}{2}ee'\cos(\lambda + \lambda' - \varpi - \varpi') \\ &+ \frac{1}{4}e'^2 + \frac{1}{4}e'^2\cos(2\lambda' - 2\varpi')\end{aligned} \tag{51}$$

We can collect the components

$$G_1 = 1, \tag{52}$$
$$G_e = -\cos(\lambda - \varpi)\alpha D, \tag{53}$$
$$G_{e'} = \cos(\lambda' - \varpi')(1 + \alpha D), \tag{54}$$
$$G_{ee'} = \left[\cos(\lambda - \lambda' - \varpi + \varpi') + \cos(\lambda + \lambda' - \varpi - \varpi')\right](-\alpha D - \alpha^2 D^2/2), \tag{55}$$
$$G_{e^2} = \frac{1}{2}\left[1 - \cos(2\lambda - 2\varpi)\right]\alpha D + \frac{1}{4}\left[1 + \cos(2\lambda - 2\varpi)\right]\alpha^2 D^2, \tag{56}$$
$$G_{e'^2} = \cos(2\lambda' - 2\omega') + \left[\frac{1}{2} + \frac{3}{2}\cos(2\lambda' - 2\varpi')\right]\alpha D + \frac{1}{4}[1 + \cos(2\lambda' - 2\varpi')]\alpha^2 D^2 \tag{57}$$

and obtain the following expansion for the operator

$$\begin{aligned}G &= 1 \\ &- e\cos(\lambda - \varpi)\alpha D \\ &+ e'\cos(\lambda' - \varpi')(1 + \alpha D) \\ &+ ee'\left[\cos(\lambda - \lambda' - \varpi + \varpi') + \cos(\lambda + \lambda' - \varpi - \varpi')\right](-\alpha D - \alpha^2 D^2/2) \\ &+ e^2\left\{\frac{1}{2}\left[1 - \cos(2\lambda - 2\varpi)\right]\alpha D + \frac{1}{4}\left[1 + \cos(2\lambda - 2\varpi)\right]\alpha^2 D^2\right\} \\ &+ e'^2\left\{\cos(2\lambda' - 2\omega') + \left[\frac{1}{2} + \frac{3}{2}\cos(2\lambda' - 2\varpi')\right]\alpha D + \frac{1}{4}[1 + \cos(2\lambda' - 2\varpi')]\alpha^2 D^2\right\}.\end{aligned} \tag{58}$$

1.2.3 Expansion of the Inclination Term

We wish to expand the inclination term in (42)

$$\delta = \cos\psi - \cos(\theta - \theta'). \tag{59}$$

To derive an expression for $\cos\psi$, we first expand the position of a planet

$$X/r = \cos\Omega\cos(\omega + f) - \sin\Omega\sin(\omega + f)\cos I, \tag{60}$$
$$Y/r = \sin\Omega\cos(\omega + f) + \cos\Omega\sin(\omega + f)\cos I, \tag{61}$$
$$Z/r = \sin(\omega + f)\sin I. \tag{62}$$

We will write $s := \sin\frac{I}{2}$, where s is considered $\mathcal{O}(e)$. Then,

$$\cos I = 1 - 2\sin^2\frac{I}{2} = 1 - 2s^2 \tag{63}$$
$$\sin I = 2\sin\frac{I}{2}\cos\frac{I}{2} = 2\sin\frac{I}{2}\sqrt{1 - \sin^2\frac{I}{2}} \approx 2s \tag{64}$$

Then,

$$\cos\psi \;=\; \frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{rr'} = \frac{XX'+YY'+ZZ'}{rr'}. \tag{65}$$

To compute this expression, we need second order expansions for $\cos(\omega+f)$ and $\sin(\omega+f)$ in terms of M and e. We will start from the expansions

$$\sin f \;=\; \sin M + e\sin 2M + e^2/8(9\sin 3M - 7\sin M) + \mathcal{O}(e^3), \tag{66}$$
$$\cos f \;=\; \cos M + e(\cos 2M - 1) + e^2/8(9\cos 3M - 9\cos M) + \mathcal{O}(e^3). \tag{67}$$

Then,

$$\begin{aligned}
\cos(\omega+f) &= \cos\omega\cos f - \sin\omega\sin f && (68)\\
&= \cos\omega[\cos M + e(\cos 2M - 1) + \frac{e^2}{8}(9\cos 3M - 9\cos M)] && (69)\\
&- \sin\omega[\sin M + e\sin 2M + \frac{e^2}{8}(9\sin 3M - 7\sin M)]\\
&= \cos\omega\cos M - \sin\omega\sin M && (70)\\
&+ e(\cos\omega\cos 2M - \sin\omega\sin 2M - \cos\omega)\\
&+ \frac{e^2}{8}(9\cos\omega\cos 3M - 9\sin\omega\sin 3M - 9\cos\omega\cos M + 7\sin\omega\sin M)
\end{aligned}$$

We can substitute

$$9\cos\omega\cos M - 7\sin\omega\sin M = 8(\cos\omega\cos M - 8\sin\omega\sin M) + (\cos\omega\cos M + \sin\omega\sin M) \tag{71}$$

to obtain

$$\begin{aligned}
\cos(\omega+f) &= \cos(\omega+M) && (72)\\
&+ e[\cos(\omega+2M) - \cos\omega]\\
&+ \frac{e^2}{8}\left[9\cos(\omega+3M) - 8\cos(\omega+M) - \cos(\omega-M)\right].
\end{aligned}$$

Similarly

$$\begin{aligned}
\sin(\omega+f) &= \sin\omega\cos f + \cos\omega\sin f && (73)\\
&= \sin\omega[\cos M + e(\cos 2M - 1) + e^2/8(9\cos 3M - 9\cos M)] && (74)\\
&+ \cos\omega[\sin M + e\sin 2M + e^2/8(9\sin 3M - 7\sin M)]\\
&= \sin\omega\cos M + \cos\omega\sin M && (75)\\
&+ e\left(\sin\omega\cos 2M + \cos\omega\sin 2M - \sin\omega\right)\\
&+ \frac{e^2}{8}\left[9\sin\omega\cos 3M + 9\cos\omega\sin 3M - 9\sin\omega\cos M - 7\cos\omega\sin M\right]
\end{aligned}$$

We can substitute

$$9\sin\omega\cos M + 7\cos\omega\sin M = 8(\sin\omega\cos M + \cos\omega\sin M) + \sin\omega\cos M - \cos\omega\sin M \tag{76}$$

to obtain

$$\begin{aligned} \sin(\omega + f) &= \sin(\omega + M) \\ &+ e[\sin(\omega + 2M) - \sin\omega] \\ &+ \frac{e^2}{8}\left[9\sin(\omega + 3M) - 8\sin(\omega + M) - \sin(\omega - M)\right]. \end{aligned} \tag{77}$$

The expressions $X/r, Y/r, Z/r$ now follow

$$
\begin{aligned}
(X/r)_1 &= \cos\Omega\cos(\omega+f)_1 - \sin\Omega\sin(\omega+f)_1(\cos I)_1 && (78)\\
&= \cos\Omega\cos(\omega+M) - \sin\Omega\sin(\omega+M) && (79)\\
&= \cos(\Omega+\omega+M), && (80)\\
(Y/r)_1 &= \sin\Omega\cos(\omega+f)_1 + \cos\Omega\sin(\omega+f)_1(\cos I)_1 && (81)\\
&= \sin\Omega\cos(\omega+M) + \cos\Omega\sin(\omega+M) && (82)\\
&= \sin(\Omega+\omega+M), && (83)\\
(Z/r)_1 &= \sin(\omega+f)_1(\sin I)_1 && (84)\\
&= 0, && (85)\\
(X/r)_s &= 0, && (86)\\
(Y/r)_s &= 0, && (87)\\
(Z/r)_s &= \sin(\omega+f)_1(\sin I)_s && (88)\\
&= 2\sin(\omega+M), && (89)\\
(X/r)_{s^2} &= -\sin\Omega\sin(\omega+f)_1(\cos I)_{s^2} && (90)\\
&= 2\sin\Omega\sin(\omega+M) && (91)\\
&= \cos(\Omega-\omega-M) - \cos(\Omega+\omega+M), && (92)\\
(Y/r)_{s^2} &= \cos\Omega\sin(\omega+f)_1(\cos I)_{s^2} && (93)\\
&= -2\cos\Omega\sin(\omega+M) && (94)\\
&= \sin(\Omega-\omega-M) - \sin(\Omega+\omega+M), && (95)\\
(Z/r)_{s^2} &= 0, && (96)\\
(X/r)_e &= \cos\Omega\cos(\omega+f)_e - \sin\Omega\sin(\omega+f)_e(\cos I)_1 && (97)\\
&= \cos\Omega[\cos(\omega+2M) - \cos\omega] - \sin\Omega[\sin(\omega+2M) - \sin\omega] && (98)\\
&= \cos(\Omega+\omega+2M) - \cos(\Omega+\omega) && (99)\\
(Y/r)_e &= \sin\Omega\cos(\omega+f)_e + \cos\Omega\sin(\omega+f)_e(\cos I)_1 && (100)\\
&= \sin\Omega[\cos(\omega+2M) - \cos\omega] + \cos\Omega[\sin(\omega+2M) - \sin\omega] && (101)\\
&= \sin(\Omega+\omega+2M) - \sin(\Omega+\omega) && (102)\\
(Z/r)_e &= \sin(\omega+f)_e\sin(I)_1 && (103)\\
&= 0, && (104)\\
(X/r)_{es} &= \cos\Omega\cos(\omega+f)_e - \sin\Omega\sin(\omega+f)_e(\cos I)_s && (105)\\
&= 0, && (106)\\
(Y/r)_{es} &= \sin\Omega\cos(\omega+f)_e + \cos\Omega\sin(\omega+f)_e(\cos I)_s && (107)\\
&= 0, && (108)\\
(Z/r)_{es} &= \sin(\omega+f)_e(\sin I)_s && (109)\\
&= 2\cos(\omega+2M) - 2\cos\omega && (110)
\end{aligned}
$$

$$
\begin{aligned}
(X/r)_{e^2} &= \cos\Omega\cos(\omega+f)_{e^2} - \sin\Omega\sin(\omega+f)_{e^2}(\cos I)_1 && (111)\\
&= \frac{1}{8}\cos\Omega\,[9\cos(\omega+3M) - 8\cos(\omega+M) - \cos(\omega-M)] && (112)\\
&- \frac{1}{8}\sin\Omega\,[9\sin(\omega+3M) - 8\sin(\omega+M) - \sin(\omega-M)] \\
&= \frac{9}{8}\cos(\Omega+\omega+3M) - \cos(\Omega+\omega+M) - \frac{1}{8}\cos(\Omega+\omega-M) && (113)\\
(Y/r)_{e^2} &= \sin\Omega\cos(\omega+f)_{e^2} + \cos\Omega\sin(\omega+f)_{e^2}(\cos I)_1 && (114)\\
&= \frac{1}{8}\sin\Omega\,[9\cos(\omega+3M) - 8\cos(\omega+M) - \cos(\omega-M)] && (115)\\
&+ \frac{1}{8}\cos\Omega\,[9\sin(\omega+3M) - 8\sin(\omega+M) - \sin(\omega-M)] \\
&= \frac{9}{8}\sin(\Omega+\omega+3M) - \sin(\Omega+\omega+M) - \frac{1}{8}\sin(\Omega+\omega-M), && (116)\\
(Z/r)_{e^2} &= \sin(\omega+f)_{e^2}(\sin I)_1 && (117)\\
&= 0. && (118)
\end{aligned}
$$

Now

$$
\begin{aligned}
(\cos\psi)_1 &= (X/r)_1(X'/r')_1 + (Y/r)_1(Y'/r')_1 + (Z/r)_1(Z'/r')_1 && (119)\\
&= \cos(\Omega+\omega+M)\cos(\Omega'+\omega'+M') + \sin(\Omega+\omega+M)\sin(\Omega'+\omega'+M') && (120)\\
&= \cos(\Omega+\omega+M-\Omega'-\omega'-M') && (121)\\
&= \cos(\lambda-\lambda') && (122)\\
(\cos\psi)_s &= (Z/r)_s(Z'/r')_1 = 0 && (123)\\
(\cos\psi)_{s'} &= (Z/r)_1(Z'/r')_{s'} = 0 && (124)\\
(\cos\psi)_{ss'} &= (Z/r)_s(Z'/r')_{s'} && (125)\\
&= 4\sin(\omega+M)\sin(\omega'+M') && (126)\\
&= 2\cos(\lambda-\lambda'-\Omega+\Omega') - 2\cos(\lambda+\lambda'-\Omega-\Omega') && (127)\\
(\cos\psi)_{s^2} &= (X/r)_{s^2}(X'/r')_1 + (Y/r)_{s^2}(Y'/r')_1 && (128)\\
&= [\cos(\Omega-\omega-M) - \cos(\Omega+\omega+M)]\cos(\Omega'+\omega'+M') && (129)\\
&+ [\sin(\Omega-\omega-M) - \sin(\Omega+\omega+M)]\sin(\Omega'+\omega'+M') \\
&= \cos(\Omega-\omega-M-\Omega'-\omega'-M) - \cos(\Omega+\omega+M-\Omega'-\omega'-M) && (130)\\
&= \cos(\lambda+\lambda'-2\Omega) - \cos(\lambda-\lambda') && (131)\\
(\cos\psi)_{s'^2} &= \cos(\lambda+\lambda'-2\Omega') - \cos(\lambda-\lambda') && (132)\\
(\cos\psi)_e &= (X/r)_e(X'/r')_1 + (Y/r)_e(Y'/r')_1 && (133)\\
&= [\cos(\lambda+M) - \cos(\lambda-M)]\cos\lambda' + [\sin(\lambda+M) - \sin(\lambda-M)]\sin\lambda' && (134)\\
&= \cos(\lambda-\lambda'+M) - \cos(\lambda-\lambda'-M) && (135)\\
&= \cos(2\lambda-\lambda'-\varpi) - \cos(\lambda'-\varpi), && (136)\\
(\cos\psi)_{e'} &= \cos(2\lambda'-\lambda-\varpi') - \cos(\lambda-\varpi') && (137)\\
&= \cos(\lambda-2\lambda'+\varpi') - \cos(\lambda-\varpi'), && (138)
\end{aligned}
$$

$$
\begin{aligned}
(\cos\psi)_{es} &= (Z/r)_{es}(Z'/r')_1 = 0, && (139)\\
(\cos\psi)_{e's'} &= (Z/r)_1(Z'/r')_{e's'} = 0, && (140)\\
(\cos\psi)_{ee'} &= (X/r)_e(X'/r')_{e'} + (Y/r)_e(Y'/r')_{e'} && (141)\\
&= [\cos(\lambda+M)-\cos(\lambda-M)]\left[\cos(\lambda'+M')-\cos(\lambda'-M')\right] && (142)\\
&+ [\sin(\lambda+M)-\sin(\lambda-M)]\left[\sin(\lambda'+M')-\sin(\lambda'-M')\right]\\
&= \cos(\lambda+M-\lambda'-M')-\cos(\lambda+M-\lambda'+M') && (143)\\
&- \cos(\lambda-M-\lambda'-M')+\cos(\lambda-M-\lambda'+M')\\
&= \cos(2\lambda-2\lambda'-\varpi+\varpi')-\cos(2\lambda-\varpi-\varpi') && (144)\\
&- \cos(2\lambda'-\varpi'-\varpi)+\cos(\varpi-\varpi')\\
(\cos\psi)_{e^2} &= (X/r)_{e^2}(X'/r')_1 + (Y/r)_{e^2}(Y'/r')_1 && (145)\\
&= \left[\frac{9}{8}\cos(\Omega+\omega+3M)-\cos(\Omega+\omega+M)-\frac{1}{8}\cos(\Omega+\omega-M)\right]\cos\lambda' && (146)\\
&+ \left[\frac{9}{8}\sin(\Omega+\omega+3M)-\sin(\Omega+\omega+M)-\frac{1}{8}\sin(\Omega+\omega-M)\right]\sin\lambda'\\
&= \frac{9}{8}\cos(\lambda+2M-\lambda')-\cos(\lambda-\lambda')-\frac{1}{8}\cos(\lambda-2M-\lambda') && (147)\\
&= \frac{9}{8}\cos(3\lambda-\lambda'-2\varpi)-\cos(\lambda-\lambda')-\frac{1}{8}\cos(\lambda+\lambda'-2\varpi), && (148)\\
(\cos\psi)_{e'^2} &= \frac{9}{8}\cos(3\lambda'-\lambda-2\varpi')-\cos(\lambda-\lambda')-\frac{1}{8}\cos(\lambda'+\lambda-2\varpi'), && (149)
\end{aligned}
$$

Collecting the terms yields

$$
\begin{aligned}
\cos\psi &= \cos(\lambda-\lambda') && (150)\\
&+ e[\cos(2\lambda-\lambda'-\varpi)-\cos(\lambda'-\varpi)]\\
&+ e'[\cos(2\lambda'-\lambda-\varpi')-\cos(\lambda-\varpi')]\\
&+ ee'[\cos(2\lambda-2\lambda'-\varpi+\varpi')-\cos(2\lambda-\varpi-\varpi')-\cos(2\lambda'-\varpi'-\varpi)+\cos(\varpi-\varpi')]\\
&+ e^2\left[\frac{9}{8}\cos(3\lambda-\lambda'-2\varpi)-\cos(\lambda-\lambda')-\frac{1}{8}\cos(\lambda+\lambda'-2\varpi)\right]\\
&+ e'^2\left[\frac{9}{8}\cos(3\lambda'-\lambda-2\varpi')-\cos(\lambda-\lambda')-\frac{1}{8}\cos(\lambda'+\lambda-2\varpi')\right]\\
&+ 2ss'\left[2\cos(\lambda-\lambda'-\Omega+\Omega')-2\cos(\lambda+\lambda'-\Omega-\Omega')\right]\\
&+ s^2\left[\cos(\lambda+\lambda'-2\Omega)-\cos(\lambda-\lambda')\right]\\
&+ s'^2\left[\cos(\lambda'+\lambda-2\Omega')-\cos(\lambda-\lambda')\right].
\end{aligned}
$$

When inclination for both planets is equal to 0, we get $s = s' = 0$ and $\cos\psi = \cos(\theta-\theta')$. Moreover inclination doesn't influence the true anomaly. Thus,

$$
\begin{aligned}
\delta &= 2ss'\left[\cos(\lambda-\lambda'-\Omega+\Omega')-\cos(\lambda+\lambda'-\Omega-\Omega')\right] && (151)\\
&+ s^2\left[\cos(\lambda+\lambda'-2\Omega)-\cos(\lambda-\lambda')\right]\\
&+ s'^2\left[\cos(\lambda'+\lambda-2\Omega')-\cos(\lambda-\lambda')\right].
\end{aligned}
$$

1.2.4 Expansion of $\cos[j(\theta - \theta')]$

We wish to express $\cos[j(\theta-\theta')]$ as a function of the mean longitudes $\lambda = \Omega+\omega+M$, $\lambda' = \Omega'+\omega'+M'$ and the longitudes of perihelion $\varpi = \Omega+\omega$, $\varpi' = \Omega'+\omega'$. At small eccentricities the difference between the true anomaly f and the mean anomaly M is small. Thus, we use the second order series expansion

$$f = M + 2e\sin M + \frac{5}{4}e^2 \sin 2M + \mathcal{O}(e^3) \tag{152}$$

and second order Taylor series expansion

$$\cos x = \cos x_0 - \sin x_0(x - x_0) - \frac{1}{2}\cos x_0(x - x_0)^2 + \mathcal{O}[(x - x_0)^3], \tag{153}$$

$$\sin x = \sin x_0 + \cos x_0(x - x_0) - \frac{1}{2}\sin x_0(x - x_0)^2 + \mathcal{O}[(x - x_0)^3], \tag{154}$$

where $x = jf$ and $x_0 = jM$. The computation is carried out as follows:

1. Second-order expansions for $\sin jf$ and $\cos jf$.
2. $\cos j\theta$ and $\sin j\theta$ using trigonometric addition formulas with ϖ.
3. $\cos[j(\theta - \theta')]$ with trigonometric subtraction formula.

That, we will start by expanding:

$$\cos jf \approx \cos jM - \sin jM[j(f - M)] - \frac{1}{2}\cos jM[j(f - M)]^2 \tag{155}$$

$$= \cos jM - \sin jM\left[j\left(2e\sin M + \frac{5}{4}e^2\sin 2M\right)\right] - \frac{1}{2}\cos jM\left[j^2(4e^2\sin^2 M)\right] \tag{156}$$

$$= \cos jM - 2ej\sin jM\sin M - e^2\left[\frac{5}{4}j\sin jM\sin 2M + j^2\cos jM(1-\cos 2M)\right], \tag{157}$$

$$\sin jf \approx \sin jM + \cos jM(f - M) - \frac{1}{2}\sin jM(f - M)^2 \tag{158}$$

$$= \sin jM + \cos jM\left[j\left(2e\sin M + \frac{5}{4}e^2\sin 2M\right)\right] - \frac{1}{2}\sin jM\left[j^2\left(4e^2\sin^2 M\right)\right] \tag{159}$$

$$= \sin jM + 2ej\cos jM\sin M + e^2\left[\frac{5}{4}j\cos jM\sin 2M - j^2\sin jM\,(1-\cos 2M)\right] \tag{160}$$

where we have used the trigonometric formula $2\sin^2 M = 1 - \cos 2M$. That is, specific orders can be written

$$(\cos jf)_1 = \cos jM, \tag{161}$$

$$(\cos jf)_e = -2j\sin jM\sin M, \tag{162}$$

$$(\cos jf)_{e^2} = -\frac{5}{4}j\sin jM\sin 2M - j^2\cos jM(1-\cos 2M), \tag{163}$$

$$(\sin jf)_1 = \sin jM, \tag{164}$$

$$(\sin jf)_e = 2j\cos jM\sin M, \tag{165}$$

$$(\sin jf)_{e^2}, = \frac{5}{4}j\cos jM\sin 2M - j^2\sin jM\,(1-\cos 2M)\,. \tag{166}$$

Let us now compute the expressions for the true anomaly

$$\cos j\theta = \cos j\varpi \cos jf - \sin j\varpi \sin jf \tag{167}$$
$$\sin j\theta = \sin j\varpi \cos jf + \cos j\varpi \sin jf. \tag{168}$$

This is most conveninent to do separately for the orders $1, e, e^2$ as follows:

$$\begin{aligned}
(\cos j\theta)_1 &= \cos j\varpi (\cos jf)_1 - \sin j\varpi (\sin jf)_1 && (169)\\
&= \cos j\varpi \cos jM - \sin j\varpi \sin jM && (170)\\
&= \cos[j(\varpi + M)] && (171)\\
&= \cos j\lambda, && (172)\\
(\sin j\theta)_1 &= \sin j\varpi (\cos jf)_1 + \cos j\varpi (\sin jf)_1 && (173)\\
&= \sin j\varpi \cos jM + \cos j\varpi \sin jM && (174)\\
&= \sin[j(\varpi + f)] && (175)\\
&= \sin j\lambda. && (176)
\end{aligned}$$

$$\begin{aligned}
(\cos j\theta)_e &= \cos j\varpi (\cos jf)_e - \sin j\varpi (\sin jf)_e && (177)\\
&= \cos j\varpi \,(-2 \sin jM \sin M) - \sin j\varpi \,(2j \cos jM \sin M) && (178)\\
&= -2j \sin M \,(\cos j\varpi \sin jM + \sin j\varpi \cos jM) && (179)\\
&= -2j \sin M \sin j\lambda && (180)\\
&= j \cos(j\lambda + M) - j \cos(j\lambda - M) && (181)\\
&= j \cos[(1+j)\lambda - \varpi] - j \cos[(1-j)\lambda + \varpi], && (182)\\
(\sin j\theta)_e &= \sin j\varpi (\cos jf)_e + \cos j\varpi (\sin jf)_e && (183)\\
&= \sin j\varpi (-2j \sin jM \sin M) + \cos j\varpi (2j \cos jM \sin M) && (184)\\
&= 2j \sin M(-\sin j\varpi \sin jM + \cos j\varpi \cos jM) && (185)\\
&= 2j \sin M \cos j\lambda && (186)\\
&= j \sin(M + j\lambda) + j \sin(M - j\lambda) && (187)\\
&= j \sin[(1+j)\lambda - \varpi] + j \sin[(1-j)\lambda - \varpi]. && (188)
\end{aligned}$$

where we have used the trigonometric identities $2 \sin x \sin y = \cos(x-y) - \cos(x+y)$ and $2 \sin x \cos y = \sin(x+y) + \sin(x-y)$.

$$\begin{aligned}
(\cos j\theta)_{e^2} &= \cos j\varpi (\cos jf)_{e^2} - \sin j\varpi (\sin jf)_{e^2} && (189)\\
&= \cos j\varpi \left[-\frac{5}{4} j \sin jM \sin 2M - j^2 \cos jM (1 - \cos 2M)\right] && (190)\\
&- \sin j\varpi \left[\frac{5}{4} j \cos jM \sin 2M - j^2 \sin jM \,(1 - \cos 2M)\right] \\
&= -\frac{5}{4} j \sin 2M \,[\cos j\varpi \sin jM + \sin j\varpi \cos jM] && (191)\\
&- j^2 \,(1 - \cos 2M)\,(\cos j\varpi \cos jM - \sin j\varpi \sin jM) \\
&= -\frac{5}{4} j \sin 2M \sin j\lambda - j^2 (1 - \cos 2M) \cos j\lambda && (192)\\
&= -j^2 \cos j\lambda - \frac{5}{4} j \sin 2M \sin j\lambda + j^2 \cos 2M \cos j\lambda && (193)
\end{aligned}$$

Substitution of

$$\sin 2M \sin j\lambda = \frac{1}{2}\cos(j\lambda - 2M) - \frac{1}{2}\cos(j\lambda + 2M) \quad (194)$$

$$= \frac{1}{2}\cos[(2-j)\lambda - 2\varpi] - \frac{1}{2}\cos[(2+j)\lambda - 2\varpi], \quad (195)$$

$$\cos 2M \cos j\lambda = \frac{1}{2}\cos(j\lambda - 2M) + \frac{1}{2}\cos(j\lambda + 2M) \quad (196)$$

$$= \frac{1}{2}\cos[(2-j)\lambda - 2\varpi] + \frac{1}{2}\cos[(2+j)\lambda - 2\varpi]. \quad (197)$$

yields

$$(\cos j\theta)_{e^2} = -j^2 \cos j\lambda + \left(\frac{1}{2}j^2 - \frac{5}{8}j\right)\cos[(2-j)\lambda - 2\varpi] + \left(\frac{1}{2}j^2 + \frac{5}{8}j\right)\cos[(2+j)\lambda - 2\varpi]. \quad (198)$$

Similarly

$$(\sin j\theta)_{e^2} = \sin j\varpi(\cos jf)_{e^2} + \cos j\varpi(\sin jf)_{e^2} \quad (199)$$

$$= \sin j\varpi\left[-\frac{5}{4}j \sin jM \sin 2M - j^2 \cos jM(1 - \cos 2M)\right] \quad (200)$$

$$+ \cos j\varpi\left[\frac{5}{4}j \cos jM \sin 2M - j^2 \sin jM\,(1 - \cos 2M)\right] \quad (201)$$

$$= -\frac{5}{4}j \sin 2M\,(\sin j\varpi \sin jM - \cos j\varpi \cos jM) \quad (202)$$

$$- j^2(1 - \cos 2M)\,(\sin j\varpi \cos jM + \cos j\varpi \sin jM)$$

$$= \frac{5}{4}j \sin 2M \cos j\lambda - j^2(1 - \cos 2M)\sin j\lambda \quad (203)$$

Substitution of

$$\sin 2M \cos j\lambda = \frac{1}{2}\sin(2M + j\lambda) + \frac{1}{2}\sin(2M - j\lambda) \quad (204)$$

$$= \frac{1}{2}\sin[(2+j)\lambda - 2\varpi] + \frac{1}{2}\sin[(2-j)\lambda - 2\varpi], \quad (205)$$

$$\cos 2M \sin j\lambda = \frac{1}{2}\sin(2M + j\lambda) - \frac{1}{2}\sin(2M - j\lambda) \quad (206)$$

$$= \frac{1}{2}\sin[(2+j)\lambda - 2\varpi] - \frac{1}{2}\sin[(2-j)\lambda - 2\varpi]. \quad (207)$$

yields

$$(\sin j\theta)_{e^2} = -j^2 \sin j\lambda + \left(\frac{5}{8}j - \frac{1}{2}j^2\right)\sin[(2-j)\lambda - 2\varpi] + \left(\frac{5}{8}j + \frac{1}{2}j^2\right)\sin[(2+j)\lambda - 2\varpi]. \quad (208)$$

That is, collecting the results

$$(\cos j\theta)_1 = \cos j\lambda, \quad (209)$$

$$(\sin j\theta)_1 = \sin j\lambda, \quad (210)$$

$$(\cos j\theta)_e = j\cos[(1+j)\lambda - \varpi] - j\cos[(1-j)\lambda - \varpi], \quad (211)$$

$$(\sin j\theta)_e = j\sin[(1+j)\lambda - \varpi] + j\sin[(1-j)\lambda - \varpi], \quad (212)$$

$$(\cos j\theta)_{e^2} = -j^2 \cos j\lambda + \left(\frac{1}{2}j^2 - \frac{5}{8}j\right)\cos[(2-j)\lambda - 2\varpi] + \left(\frac{1}{2}j^2 + \frac{5}{8}j\right)\cos[(2+j)\lambda - 2\varpi]. \quad (213)$$

$$(\sin j\theta)_{e^2} = -j^2 \sin j\lambda - \left(\frac{1}{2}j^2 - \frac{5}{8}j\right)\sin[(2-j)\lambda - 2\varpi] + \left(\frac{1}{2}j^2 + \frac{5}{8}j\right)\sin[(2+j)\lambda - 2\varpi] \quad (214)$$

allows us to expand

$$\begin{aligned}
\cos j\theta &= (1-e^2j^2)\cos j\lambda && (215)\\
&+ ej\cos[(1+j)\lambda-\varpi]-ej\cos[(1-j)\lambda-\varpi],\\
&+ e^2\left(\frac{1}{2}j^2-\frac{5}{8}j\right)\cos[(2-j)\lambda-2\varpi]+e^2\left(\frac{1}{2}j^2+\frac{5}{8}j\right)\cos[(2+j)\lambda-2\varpi].\\
\sin j\theta &= (1-e^2j^2)\sin j\lambda && (216)\\
&+ ej\sin[(1+j)\lambda-\varpi]+ej\sin[(1-j)\lambda-\varpi]\\
&- e^2\left(\frac{1}{2}j^2-\frac{5}{8}j\right)\sin[(2-j)\lambda-2\varpi]+e^2\left(\frac{1}{2}j^2+\frac{5}{8}j\right)\sin[(2+j)\lambda-2\varpi],
\end{aligned}$$

Now the desired result can be finally computed

$$\begin{aligned}
\cos[j(\theta-\theta')] &= \cos j\theta\cos j\theta'+\sin j\theta\sin j\theta' && (217)\\
&= (\cos j\theta)_1(\cos j\theta')_1+(\sin j\theta)_1(\sin j\theta')_1 && (218)\\
&+ e(\cos j\theta)_e(\cos j\theta')_1+e(\sin j\theta)_e(\sin j\theta')_1\\
&+ e'(\cos j\theta)_1(\cos j\theta')_{e'}+e'(\sin j\theta)_1(\sin j\theta')_{e'}\\
&+ ee'(\cos j\theta)_e(\cos j\theta')_{e'}+ee'(\sin j\theta)_e(\sin j\theta')_{e'}\\
&+ e^2(\cos j\theta)_{e^2}(\cos j\theta')_1+e^2(\sin j\theta)_{e^2}(\sin j\theta')_1\\
&+ e'^2(\cos j\theta)_1(\cos j\theta')_{e'^2}+e'^2(\sin j\theta)_1(\sin j\theta')_{e'^2},
\end{aligned}$$

where

$$\begin{aligned}
\cos[j(\theta-\theta')]_1 &= (\cos j\theta)_1(\cos j\theta')_1+(\sin j\theta)_1(\sin j\theta')_1 && (219)\\
&= \cos j\lambda\cos j\lambda'+\sin j\lambda\sin j\lambda' && (220)\\
&= \cos[j(\lambda-\lambda')], && (221)
\end{aligned}$$

$$\begin{aligned}
\cos[j(\theta-\theta')]_e &= (\cos j\theta)_e(\cos j\theta')_1+(\sin j\theta)_e(\sin j\theta')_1 && (222)\\
&= j\cos[(1+j)\lambda-\varpi]\cos j\lambda'-j\cos[(1-j)\lambda-\varpi]\cos j\lambda' && (223)\\
&+ j\sin[(1+j)\lambda-\varpi]\sin j\lambda'+j\sin[(1-j)\lambda-\varpi]\sin j\lambda'\\
&= j\cos[(1+j)\lambda-j\lambda'-\varpi]-j\cos[(1-j)\lambda+j\lambda'-\varpi], && (224)
\end{aligned}$$

$$\begin{aligned}
\cos[j(\theta-\theta')]_{e'} &= (\cos j\theta)_1(\cos j\theta')_{e'}+(\sin j\theta)_1(\sin j\theta')_{e'} && (225)\\
&= j\cos j\lambda\cos[(1+j)\lambda'-\varpi']-j\cos j\lambda\cos[(1-j)\lambda'-\varpi'] && (226)\\
&+ j\sin j\lambda\sin[(1+j)\lambda'-\varpi']+j\sin j\lambda\sin[(1-j)\lambda'-\varpi'],\\
&= j\cos[(1+j)\lambda'-j\lambda-\varpi']-j\cos[(1-j)\lambda'+j\lambda-\varpi'], && (227)
\end{aligned}$$

$$\cos[j(\theta-\theta')]_{ee'} \tag{228}$$

$$= (\cos j\theta)_e(\cos j\theta')_{e'} + (\sin j\theta)_e(\sin j\theta')_{e'}$$

$$= j^2\left\{\cos[(1+j)\lambda-\varpi]-\cos[(1-j)\lambda-\varpi]\right\}\left\{\cos[(1+j)\lambda'-\varpi']-\cos[(1-j)\lambda'-\varpi']\right\} \tag{229}$$

$$+ j^2\left\{\sin[(1+j)\lambda-\varpi]+\sin[(1-j)\lambda-\varpi]\right\}\left\{\sin[(1+j)\lambda'-\varpi']+\sin[(1-j)\lambda'-\varpi']\right\}$$

$$= j^2\cos[(1+j)\lambda-\varpi]\cos[(1+j)\lambda'-\varpi'] - j^2\cos[(1+j)\lambda-\varpi]\cos[(1-j)\lambda'-\varpi'] \tag{230}$$

$$- j^2\cos[(1-j)\lambda-\varpi]\cos[(1+j)\lambda'-\varpi'] + j^2\cos[(1-j)\lambda-\varpi]\cos[(1-j)\lambda'-\varpi']$$

$$+ j^2\sin[(1+j)\lambda-\varpi]\sin[(1+j)\lambda'-\varpi'] + j^2\sin[(1+j)\lambda-\varpi]\sin[(1-j)\lambda'-\varpi']$$

$$+ j^2\sin[(1-j)\lambda-\varpi]\sin[(1+j)\lambda'-\varpi'] + j^2\sin[(1-j)\lambda-\varpi]\sin[(1-j)\lambda'-\varpi']$$

$$= j^2\cos[(1+j)\lambda-(1+j)\lambda'-\varpi+\varpi'] + j^2\cos[(1-j)\lambda-(1-j)\lambda'-\varpi+\varpi'] \tag{231}$$

$$- j^2\cos[(1+j)\lambda+(1-j)\lambda'-\varpi-\varpi'] - j^2\cos[(1-j)\lambda+(1+j)\lambda'-\varpi-\varpi']$$

$$\cos[j(\theta-\theta')]_{e^2} \tag{232}$$

$$= (\cos j\theta)_{e^2}(\cos j\theta')_1 + (\sin j\theta)_{e^2}(\sin j\theta')_1$$

$$= \left[-j^2\cos j\lambda + \left(\frac{1}{2}j^2-\frac{5}{8}j\right)\cos[(2-j)\lambda-2\varpi] + \left(\frac{1}{2}j^2+\frac{5}{8}j\right)\cos[(2+j)\lambda-2\varpi]\right]\cos j\lambda' \tag{233}$$

$$+ \left[-j^2\sin j\lambda - \left(\frac{1}{2}j^2-\frac{5}{8}j\right)\sin[(2-j)\lambda-2\varpi] + \left(\frac{1}{2}j^2+\frac{5}{8}j\right)\sin[(2+j)\lambda-2\varpi]\right]\sin j\lambda'$$

$$= -j^2\left(\cos j\lambda\cos j\lambda' + \sin j\lambda\sin j\lambda'\right) \tag{234}$$

$$+ \left(\frac{1}{2}j^2-\frac{5}{8}j\right)\left\{\cos[(2-j)\lambda-2\varpi]\cos j\lambda' - \sin[(2-j)\lambda-2\varpi]\sin j\lambda'\right\}$$

$$+ \left(\frac{1}{2}j^2+\frac{5}{8}j\right)\left\{\cos[(2+j)\lambda-2\varpi]\cos j\lambda' + \sin[(2+j)\lambda-2\varpi]\sin j\lambda'\right\}$$

$$= -j^2\cos[j(\lambda-\lambda')] \tag{235}$$

$$+ \left(\frac{1}{2}j^2-\frac{5}{8}j\right)\cos[(2-j)\lambda+j\lambda'-2\varpi] + \left(\frac{1}{2}j^2+\frac{5}{8}j\right)\cos[(2+j)\lambda-j\lambda'-2\varpi]$$

The e'^2 case follows by symmetry. Thus, we can again collect the terms

$$\begin{aligned}
\cos[j(\theta-\theta')]_1 &= \cos[j(\lambda-\lambda')], && (236)\\
\cos[j(\theta-\theta')]_e &= j\cos[(1+j)\lambda - j\lambda' - \varpi] - j\cos[(1-j)\lambda + j\lambda' - \varpi], && (237)\\
\cos[j(\theta-\theta')]_{e'} &= j\cos[(1+j)\lambda' - j\lambda - \varpi'] - j\cos[(1-j)\lambda' + j\lambda - \varpi'], && (238)\\
\cos[j(\theta-\theta')]_{ee'} &= j^2\cos[(1+j)\lambda - (1+j)\lambda' - \varpi + \varpi'] && (239)\\
&+ j^2\cos[(1-j)\lambda - (1-j)\lambda' - \varpi + \varpi']\\
&- j^2\cos[(1+j)\lambda + (1-j)\lambda' - \varpi - \varpi']\\
&- j^2\cos[(1-j)\lambda + (1+j)\lambda' - \varpi - \varpi']\\
\cos[j(\theta-\theta')]_{e^2} &= -j^2\cos[j(\lambda-\lambda')] && (240)\\
&+ \left(\frac{1}{2}j^2 - \frac{5}{8}j\right)\cos[(2-j)\lambda + j\lambda' - 2\varpi]\\
&+ \left(\frac{1}{2}j^2 + \frac{5}{8}j\right)\cos[(2+j)\lambda - j\lambda' - 2\varpi]\\
\cos[j(\theta-\theta')]_{e'^2} &= -j^2\cos[j(\lambda-\lambda')] && (241)\\
&+ \left(\frac{1}{2}j^2 - \frac{5}{8}j\right)\cos[(2-j)\lambda' + j\lambda - 2\varpi']\\
&+ \left(\frac{1}{2}j^2 + \frac{5}{8}j\right)\cos[(2+j)\lambda' - j\lambda - 2\varpi']
\end{aligned}$$

to finally expand

$$\begin{aligned}
\cos[j(\theta-\theta')] &= (1 - e^2j^2 - e'^2j^2)\cos[j(\lambda-\lambda')] && (242)\\
&+ ej\cos[(1+j)\lambda - j\lambda' - \varpi] - ej\cos[(1-j)\lambda + j\lambda' - \varpi]\\
&+ e'j\cos[(1+j)\lambda' - j\lambda - \varpi'] - e'j\cos[(1-j)\lambda' + j\lambda - \varpi']\\
&+ ee'j^2\cos[(1+j)\lambda - (1+j)\lambda' - \varpi + \varpi']\\
&+ ee'j^2\cos[(1-j)\lambda - (1-j)\lambda' - \varpi + \varpi']\\
&- ee'j^2\cos[(1+j)\lambda + (1-j)\lambda' - \varpi - \varpi']\\
&- ee'j^2\cos[(1-j)\lambda + (1+j)\lambda' - \varpi - \varpi']\\
&+ e^2\left(\frac{1}{2}j^2 - \frac{5}{8}j\right)\cos[(2-j)\lambda + j\lambda' - 2\varpi]\\
&+ e^2\left(\frac{1}{2}j^2 + \frac{5}{8}j\right)\cos[(2+j)\lambda - j\lambda' - 2\varpi]\\
&+ e'^2\left(\frac{1}{2}j^2 - \frac{5}{8}j\right)\cos[(2-j)\lambda' + j\lambda - 2\varpi']\\
&+ e'^2\left(\frac{1}{2}j^2 + \frac{5}{8}j\right)\cos[(2+j)\lambda' - j\lambda - 2\varpi']
\end{aligned}$$

1.2.5 Further Expansion of S_D

Now we must compute the product of (58) and (242). This is rather formidable task. However, since most of the terms are of second order and higher-order terms are discarded, the number of product terms is significantly reduced. We have both $\cos[j(\theta-\theta')]_1 = \cos[j(\lambda-\lambda')]$ and $G_1 = 1$. Thus, multiplications

involving a second-order term are trivial. For order 1, the computation is trivial:

$$\left\{G\cos[j(\theta-\theta')]\right\}_1 \quad = \quad \cos[j(\lambda-\lambda')]. \tag{243}$$

Order e:

$$\left\{G\cos[j(\theta-\theta')]\right\}_e = G_e\cos[j(\theta-\theta')]_1 + G_1\cos[j(\theta-\theta')]_e \tag{244}$$

$$\begin{aligned} G_e\cos[j(\theta-\theta')]_1 &= -\cos(\lambda-\varpi)\alpha D\cos[j(\lambda-\lambda')] && (245)\\ &= \frac{1}{8}(-4\alpha D)\cos((1+j)\lambda - j\lambda' - \varpi) && (246)\\ &+ \frac{1}{8}(-4\alpha D)\cos((1-j)\lambda + j\lambda' - \varpi) \end{aligned}$$

$$\begin{aligned} G_1\cos[j(\theta-\theta')]_e &= j\cos[(1+j)\lambda - j\lambda' - \varpi] - j\cos[(1-j)\lambda + j\lambda' - \varpi] && (247)\\ &= \frac{1}{8}(8j)\cos[(1+j)\lambda - j\lambda' - \varpi] && (248)\\ &+ \frac{1}{8}(-8j)\cos[(1-j)\lambda + j\lambda' - \varpi] \end{aligned}$$

$$\begin{aligned} \therefore \left\{G\cos[j(\theta-\theta')]\right\}_e &= \frac{1}{8}(8j-4\alpha D)\cos((1+j)\lambda - j\lambda' - \varpi) && (249)\\ &+ \frac{1}{8}(-8j-4\alpha D)\cos((1-j)\lambda + j\lambda' - \varpi) \end{aligned}$$

Order e':

$$\left\{G\cos[j(\theta-\theta')]\right\}_{e'} = G_{e'}\cos[j(\theta-\theta')]_1 + G_1\cos[j(\theta-\theta')]_{e'} \tag{250}$$

$$\begin{aligned} G_{e'}\cos[j(\theta-\theta')]_1 &= \cos(\lambda'-\varpi')(1+\alpha D)\cos[j(\lambda-\lambda')] && (251)\\ &= \frac{1}{8}(4+4\alpha D)\cos[(1+j)\lambda' - j\lambda - \varpi'] && (252)\\ &+ \frac{1}{8}(4+4\alpha D)\cos[(1-j)\lambda' + j\lambda - \varpi'] \end{aligned}$$

$$\begin{aligned} G_1\cos[j(\theta-\theta')]_{e'} &= j\cos[(1+j)\lambda' - j\lambda - \varpi'] - j\cos[(1-j)\lambda' + j\lambda - \varpi'] && (253)\\ &= \frac{1}{8}(8j)\cos[(1+j)\lambda' - j\lambda - \varpi'] && (254)\\ &- \frac{1}{8}(8j)\cos[(1-j)\lambda' + j\lambda - \varpi']. \end{aligned}$$

$$\begin{aligned} \therefore \left\{G\cos[j(\theta-\theta')]\right\}_{e'} &= \frac{1}{8}(4+8j+4\alpha D)\cos[(1+j)\lambda' - j\lambda - \varpi'] && (255)\\ &+ \frac{1}{8}(4-8j+4\alpha D)\cos[(1-j)\lambda' + j\lambda - \varpi'] \end{aligned}$$

Order ee':

$$\begin{aligned}\left\{G\cos[j(\theta-\theta')]\right\}_{ee'} &= G_e\cos[j(\theta-\theta')]_{e'} + G_{e'}\cos[j(\theta-\theta')]_e. \\ &+ G_{e'e'}\cos[j(\theta-\theta')]_1. + G_1\cos[j(\theta-\theta')]_{ee'}.\end{aligned} \tag{256}$$

$$G_e\cos[j(\theta-\theta')]_{e'} \tag{257}$$

$$\begin{aligned}&= -\cos(\lambda-\varpi)\alpha D\left\{j\cos[(1+j)\lambda'-j\lambda-\varpi'] - j\cos[(1-j)\lambda'+j\lambda-\varpi']\right\} \\ &= (1/8)(-4j\alpha D)\cos[(1-j)\lambda+(1+j)\lambda'-\varpi-\varpi'] \qquad (258)\\ &+ (1/8)(-4j\alpha D)\cos[(1+j)\lambda-(1+j)\lambda'-\varpi+\varpi'] \\ &+ (1/8)(4j\alpha D)\cos[(1+j)\lambda+(1-j)\lambda'-\varpi-\varpi'] \\ &+ (1/8)(4j\alpha D)\cos[(1-j)\lambda-(1-j)\lambda'-\varpi+\varpi']\end{aligned}$$

$$G_{e'}\cos[j(\theta-\theta')]_e \tag{259}$$

$$\begin{aligned}&= \cos(\lambda'-\varpi')(1+\alpha D)\left\{j\cos[(1+j)\lambda-j\lambda'-\varpi] - j\cos[(1-j)\lambda+j\lambda'-\varpi]\right\} \\ &= (1/8)(4j+4j\alpha D)\cos[(1+j)\lambda+(1-j)\lambda'-\varpi-\varpi'] \qquad (260)\\ &+ (1/8)(4j+4j\alpha D)\cos[(1+j)\lambda-(1+j)\lambda'-\varpi+\varpi'] \\ &+ (1/8)(-4j-4j\alpha D)\cos[(1-j)\lambda+(1+j)\lambda'-\varpi-\varpi'] \\ &+ (1/8)(-4j-4j\alpha D)\cos[(1-j)\lambda-(1-j)\lambda'-\varpi+\varpi'].\end{aligned}$$

$$G_{e'e'}\cos[j(\theta-\theta')]_1 \tag{261}$$

$$\begin{aligned}&= \left[\cos(\lambda-\lambda'-\varpi+\varpi') + \cos(\lambda+\lambda'-\varpi-\varpi')\right](-\alpha D-\alpha^2D^2/2)\cos[j(\lambda-\lambda')] \\ &= (1/8)\left(-4\alpha D-2\alpha^2D^2\right)\cos[(1+j)\lambda-(1+j)\lambda'-\varpi+\varpi'] \qquad (262)\\ &+ (1/8)\left(-4\alpha D-2\alpha^2D^2\right)\cos[(1-j)\lambda-(1-j)\lambda'-\varpi+\varpi'] \\ &+ (1/8)\left(-4\alpha D-2\alpha^2D^2\right)\cos[(1+j)\lambda+(1-j)\lambda'-\varpi-\varpi'] \\ &+ (1/8)\left(-4\alpha D-2\alpha^2D^2\right)\cos[(1-j)\lambda+(1+j)\lambda'-\varpi-\varpi']\end{aligned}$$

$$\begin{aligned}G_1\cos[j(\theta-\theta')]_{ee'} &= (1/8)(8j^2)\cos[(1+j)\lambda-(1+j)\lambda'-\varpi+\varpi'] \\ &+ (1/8)(8j^2)\cos[(1-j)\lambda-(1-j)\lambda'-\varpi+\varpi'] \\ &+ (1/8)(-8j^2)\cos[(1+j)\lambda+(1-j)\lambda'-\varpi-\varpi'] \\ &+ (1/8)(-8j^2)\cos[(1-j)\lambda+(1+j)\lambda'-\varpi-\varpi']\end{aligned} \tag{263}$$

$$\begin{aligned}\therefore \left\{G\cos[j(\theta-\theta')]\right\}_{ee'} &= \frac{1}{8}\left(4j+8j^2-4\alpha D-2\alpha^2D^2\right)\cos[(1+j)\lambda-(1+j)\lambda'-\varpi+\varpi'] \\ &+ \frac{1}{8}\left(-4j+8j^2-4\alpha D-2\alpha^2D^2\right)\cos[(1-j)\lambda-(1-j)\lambda'-\varpi+\varpi'] \\ &+ \frac{1}{8}\left(4j-8j^2-4\alpha D+8j\alpha D-2\alpha^2D^2\right)\cos[(1+j)\lambda+(1-j)\lambda'-\varpi-\varpi'] \\ &+ \frac{1}{8}\left(-4j-8j^2-4\alpha D-8j\alpha D-2\alpha^2D^2\right)\cos[(1-j)\lambda+(1+j)\lambda'-\varpi-\varpi'].\end{aligned} \tag{264}$$

Order e^2:

$$\left\{G\cos[j(\theta-\theta')]\right\}_{e^2} = G_{e^2}\cos[j(\theta-\theta')]_1 + G_1\cos[j(\theta-\theta')]_{e^2} + G_e\cos[j(\theta-\theta')]_e. \tag{265}$$

$$\begin{aligned}
G_{e^2}\cos[j(\theta-\theta')]_1 &= \left\{\frac{1}{2}\left[1-\cos(2\lambda-2\varpi)\right]\alpha D + \frac{1}{4}\left[1+\cos(2\lambda-2\varpi)\right]\alpha^2 D^2\right\}\cos[j(\lambda-\lambda')] \\
&= \frac{1}{8}\left(4\alpha D + 2\alpha^2 D^2\right)\cos[j(\lambda-\lambda')] \qquad (266)\\
&+ \frac{1}{8}\left(-2\alpha D + \alpha^2 D^2\right)\cos[(2+j)\lambda - j\lambda' - 2\varpi] \\
&+ \frac{1}{8}\left(-2\alpha D + \alpha^2 D^2\right)\cos[(2-j)\lambda + j\lambda' - 2\varpi]
\end{aligned}$$

$$\begin{aligned}
G_1\cos[j(\theta-\theta')]_{e^2} &= \frac{1}{8}(-8j^2)\cos[j(\lambda-\lambda')] \qquad (267)\\
&+ \frac{1}{8}\left(4j^2-5j\right)\cos[(2-j)\lambda + j\lambda' - 2\varpi] \\
&+ \frac{1}{8}\left(4j^2+5j\right)\cos[(2+j)\lambda - j\lambda' - 2\varpi]
\end{aligned}$$

$$\begin{aligned}
G_e\cos[j(\theta-\theta')]_e &= \left[-\cos(\lambda-\varpi)\alpha D\right]\left\{j\cos[(1+j)\lambda - j\lambda' - \varpi] - j\cos[(1-j)\lambda + j\lambda' - \varpi]\right\} \\
&= \frac{1}{8}(-4j\alpha D)\cos[(2+j)\lambda - j\lambda' - 2\varpi] \qquad (268)\\
&+ \frac{1}{8}(-4j\alpha D)\cos[j(\lambda - j\lambda')] \\
&+ \frac{1}{8}(4j\alpha D)\cos[(2-j)\lambda + j\lambda' - \varpi)] \\
&+ \frac{1}{8}(4j\alpha D)\cos[j(\lambda - j\lambda')] \\
&= \frac{1}{8}(-4j\alpha D)\cos[(2+j)\lambda - j\lambda' - 2\varpi] \qquad (269)\\
&+ \frac{1}{8}(4j\alpha D)\cos[(2-j)\lambda + j\lambda' - \varpi)]
\end{aligned}$$

$$\begin{aligned}
\therefore \left\{G\cos[j(\theta-\theta')]\right\}_{e^2} &= \frac{1}{8}\left(-8j^2 + 4\alpha D + 2\alpha^2 D^2\right)\cos[j(\lambda-\lambda')] \qquad (270)\\
&+ \frac{1}{8}\left(5j + 4j^2 - 2\alpha D - 4j\alpha D + \alpha^2 D^2\right)\cos[(2+j)\lambda - j\lambda' - 2\varpi] \\
&+ \frac{1}{8}\left(-5j + 4j^2 - 2\alpha D + 4j\alpha D + \alpha^2 D^2\right)\cos[(2-j)\lambda + j\lambda' - 2\varpi]
\end{aligned}$$

Order e'^2:

$$\left\{G\cos[j(\theta-\theta')]\right\}_{e'^2} = G_{e'^2}\cos[j(\theta-\theta')]_1 + G_1\cos[j(\theta-\theta')]_{e'^2} + G_{e'}\cos[j(\theta-\theta')]_{e'}. \tag{271}$$

$$\begin{aligned}
& G_{e'^2}\cos[j(\theta-\theta')]_1 && (272)\\
= \;& \left\{\cos(2\lambda'-2\omega') + \left[\frac{1}{2}+\frac{3}{2}\cos(2\lambda'-2\varpi')\right]\alpha D + \frac{1}{4}[1+\cos(2\lambda'-2\varpi')]\alpha^2D^2\right\}\cos[j(\lambda-\lambda')] \\
= \;& \frac{1}{8}\left(4\alpha D + 2\alpha^2D^2\right)\cos[j(\lambda-\lambda')] && (273)\\
+ \;& \frac{1}{8}\left(4+6\alpha D+\alpha^2D^2\right)\cos[(2-j)\lambda' + j\lambda - 2\varpi'] \\
+ \;& \frac{1}{8}\left(4+6\alpha D+\alpha^2D^2\right)\cos[(2+j)\lambda' - j\lambda - 2\varpi']
\end{aligned}$$

$$\begin{aligned}
G_1\cos[j(\theta-\theta')]_{e'^2} &= -j^2\cos[j(\lambda-\lambda')] && (274)\\
&+ \left(\frac{1}{2}j^2 - \frac{5}{8}j\right)\cos[(2-j)\lambda' + j\lambda - 2\varpi'] \\
&+ \left(\frac{1}{2}j^2 + \frac{5}{8}j\right)\cos[(2+j)\lambda' - j\lambda - 2\varpi'] \\
&= \frac{1}{8}\left(-8j^2\right)\cos[j(\lambda-\lambda')] && (275)\\
&+ \frac{1}{8}\left(4j^2-5j\right)\cos[(2-j)\lambda' + j\lambda - 2\varpi'] \\
&+ \frac{1}{8}\left(4j^2+5j\right)\cos[(2+j)\lambda' - j\lambda - 2\varpi']
\end{aligned}$$

$$\begin{aligned}
& G_{e'}\cos[j(\theta-\theta)]e' && (276)\\
= \;& \cos(\lambda'-\varpi')(1+\alpha D)\left\{j\cos[(1+j)\lambda' - j\lambda - \varpi'] - j\cos[(1-j)\lambda' + j\lambda - \varpi']\right\} \\
= \;& \frac{1}{8}(4j+4j\alpha D)\cos[(2+j)\lambda' - j\lambda - 2\varpi'] + \frac{1}{8}(4j+4j\alpha D)\cos[j(\lambda-\lambda')] && (277)\\
+ \;& \frac{1}{8}(-4j-4j\alpha D)\cos[(2-j)\lambda' + j\lambda - 2\varpi'] + \frac{1}{8}(-4j-4j\alpha D)\cos[(j(\lambda-\lambda')] \\
= \;& \frac{1}{8}(4j+4j\alpha D)\cos[(2+j)\lambda' - j\lambda - 2\varpi'] && (278)\\
+ \;& \frac{1}{8}(-4j-4j\alpha D)\cos[(2-j)\lambda' + j\lambda - 2\varpi']
\end{aligned}$$

$$\begin{aligned}
\therefore \left\{G\cos[j(\theta-\theta')]\right\}_{e'^2} &= \frac{1}{8}\left(-8j^2+4\alpha D+2\alpha^2D^2\right)\cos[j(\lambda-\lambda')] && (279)\\
&+ \frac{1}{8}\left(4-9j+4j^2+6\alpha D-4j\alpha D+\alpha^2D^2\right)\cos[(2-j)\lambda' + j\lambda - 2\varpi'] \\
&+ \frac{1}{8}\left(4+9j+4j^2+6\alpha D+4j\alpha D+\alpha^2D^2\right)\cos[(2+j)\lambda' - j\lambda - 2\varpi']
\end{aligned}$$

Order ss':

$$\begin{aligned}
\left\{\delta\alpha\cos[j(\theta-\theta')]\right\}_{ss'} &= \delta_{ss'}\alpha\cos[j(\theta-\theta')]_1 && (280)\\
&= 2\alpha\left[\cos(\lambda-\lambda'-\Omega+\Omega')-\cos(\lambda+\lambda'-\Omega-\Omega')\right]\cos[j(\lambda-\lambda')] && (281)\\
&= \alpha\cos[(1-j)\lambda-(1-j)\lambda'-\Omega+\Omega'] && (282)\\
&+ \alpha\cos[(1+j)\lambda-(1+j)\lambda'-\Omega+\Omega']\\
&- \alpha\cos[(1-j)\lambda+(1+j)\lambda'-\Omega-\Omega']\\
&- \alpha\cos[(1+j)\lambda+(1-j)\lambda'-\Omega-\Omega']
\end{aligned}$$

Order s^2:

$$\begin{aligned}
\left\{\delta\alpha\cos[j(\theta-\theta')]\right\}_{s^2} &= \delta_{s^2}\alpha\cos[j(\theta-\theta')]_1 && (283)\\
&= \alpha[\cos(\lambda+\lambda'-2\Omega)-\cos(\lambda-\lambda')]\cos[j(\lambda-\lambda')] && (284)\\
&= \frac{\alpha}{2}\cos[(1+j)\lambda+(1-j)\lambda'-2\Omega] && (285)\\
&+ \frac{\alpha}{2}\cos[(1-j)\lambda+(1+j)\lambda'-2\Omega]\\
&- \frac{\alpha}{2}\cos[(1+j)\lambda-(1+j)\lambda']\\
&- \frac{\alpha}{2}\cos[(1-j)\lambda+(1-j)\lambda'].
\end{aligned}$$

Order s'^2:

$$\begin{aligned}
\left\{\delta\alpha\cos[j(\theta-\theta')]\right\}_{s'^2} &= \delta_{s'^2}\alpha\cos[j(\theta-\theta')]_1 && (286)\\
&= \alpha[\cos(\lambda'+\lambda-2\Omega')-\cos(\lambda-\lambda')]\cos[j(\lambda-\lambda')] && (287)\\
&= \frac{\alpha}{2}\cos[(1+j)\lambda'+(1-j)\lambda-2\Omega'] && (288)\\
&+ \frac{\alpha}{2}\cos[(1-j)\lambda'+(1+j)\lambda-2\Omega']\\
&- \frac{\alpha}{2}\cos[(1+j)\lambda-(1+j)\lambda']\\
&- \frac{\alpha}{2}\cos[(1-j)\lambda+(1-j)\lambda'].
\end{aligned}$$

If we now expand (42)

$$\mathcal{S}_D = \sum_{j=-\infty}^{\infty}\mathcal{S}_D^{(j)}, \tag{289}$$

we can express the components of individual terms:

$$\left(\mathcal{S}_D^{(j)}\right)_1 = \frac{1}{2}\cos[j(\lambda-\lambda')]b_{1/2}^{(j)}, \tag{290}$$

$$\begin{aligned}\left(\mathcal{S}_D^{(j)}\right)_e &= (1/4)\,(2j-1\alpha D)\cos((1+j)\lambda - j\lambda' - \varpi)b_{1/2}^{(j)} \\ &+ (1/4)\,(-2j-1\alpha D)\cos((1-j)\lambda + j\lambda' - \varpi)b_{1/2}^{(j)},\end{aligned} \tag{291}$$

$$\begin{aligned}\left(\mathcal{S}_D^{(j)}\right)_{e'} &= (1/4)\,(1+2j+\alpha D)\cos[(1+j)\lambda' - j\lambda - \varpi']b_{1/2}^{(j)} \\ &+ (1/4)\,(1-2j+\alpha D)\cos[(1-j)\lambda' + j\lambda - \varpi']b_{1/2}^{(j)},\end{aligned} \tag{292}$$

$$\begin{aligned}\left(\mathcal{S}_D^{(j)}\right)_{ee'} &= (1/8)\left(2j+4j^2-2\alpha D-\alpha^2D^2\right)\cos[(1+j)\lambda-(1+j)\lambda'-\varpi+\varpi']b_{1/2}^{(j)} \\ &+ (1/8)\left(-2j+4j^2-2\alpha D-\alpha^2D^2\right)\cos[(1-j)\lambda-(1-j)\lambda'-\varpi+\varpi']b_{1/2}^{(j)} \\ &+ (1/8)\left(2j-4j^2-2\alpha D+4j\alpha D-\alpha^2D^2\right)\cos[(1+j)\lambda+(1-j)\lambda'-\varpi-\varpi']b_{1/2}^{(j)} \\ &+ (1/8)\left(-2j-4j^2-2\alpha D-4j\alpha D-\alpha^2D^2\right)\cos[(1-j)\lambda+(1+j)\lambda'-\varpi-\varpi']b_{1/2}^{(j)},\end{aligned} \tag{293}$$

$$\begin{aligned}\left(\mathcal{S}_D^{(j)}\right)_{e^2} &= (1/8)\left(-4j^2+2\alpha D+\alpha^2D^2\right)\cos[j(\lambda-\lambda')]b_{1/2}^{(j)} \\ &+ (1/16)\left(5j+4j^2-2\alpha D-4j\alpha D+\alpha^2D^2\right)\cos[(2+j)\lambda-j\lambda'-2\varpi]b_{1/2}^{(j)} \\ &+ (1/16)\left(-5j+4j^2-2\alpha D+4j\alpha D+\alpha^2D^2\right)\cos[(2-j)\lambda+j\lambda'-2\varpi]b_{1/2}^{(j)},\end{aligned} \tag{294}$$

$$\begin{aligned}\left(\mathcal{S}_D^{(j)}\right)_{e'^2} &= (1/8)\left(-4j^2+2\alpha D+\alpha^2D^2\right)\cos[j(\lambda-\lambda')]b_{1/2}^{(j)} \\ &+ (1/16)\left(4-9j+4j^2+6\alpha D-4j\alpha D+\alpha^2D^2\right)\cos[(2-j)\lambda'+j\lambda-2\varpi']b_{1/2}^{(j)} \\ &+ (1/16)\left(4+9j+4j^2+6\alpha D+4j\alpha D+\alpha^2D^2\right)\cos[(2+j)\lambda'-j\lambda-2\varpi']b_{1/2}^{(j)},\end{aligned} \tag{295}$$

$$\begin{aligned}\left(\mathcal{S}_D^{(j)}\right)_{ss'} &= (1/2)\alpha\cos[(1-j)\lambda-(1-j)\lambda'-\Omega+\Omega']b_{3/2}^{(j)} \\ &+ (1/2)\alpha\cos[(1+j)\lambda-(1+j)\lambda'-\Omega+\Omega']b_{3/2}^{(j)} \\ &- (1/2)\alpha\cos[(1-j)\lambda+(1+j)\lambda'-\Omega-\Omega']b_{3/2}^{(j)} \\ &- (1/2)\alpha\cos[(1+j)\lambda+(1-j)\lambda'-\Omega-\Omega']b_{3/2}^{(j)},\end{aligned} \tag{296}$$

$$\begin{aligned}\left(\mathcal{S}_D^{(j)}\right)_{s^2} &= (1/4)\alpha\cos[(1+j)\lambda+(1-j)\lambda'-2\Omega]b_{3/2}^{(j)} \\ &+ (1/4)\alpha\cos[(1-j)\lambda+(1+j)\lambda'-2\Omega]b_{3/2}^{(j)} \\ &- (1/4)\alpha\cos[(1+j)\lambda-(1+j)\lambda']b_{3/2}^{(j)} \\ &- (1/4)\alpha\cos[(1-j)\lambda+(1-j)\lambda']b_{3/2}^{(j)},\end{aligned} \tag{297}$$

$$\begin{aligned}\left(\mathcal{S}_D^{(j)}\right)_{s'^2} &= (1/4)\alpha\cos[(1+j)\lambda'+(1-j)\lambda-2\Omega']b_{3/2}^{(j)} \\ &+ (1/4)\alpha\cos[(1-j)\lambda'+(1+j)\lambda-2\Omega']b_{3/2}^{(j)} \\ &- (1/4)\alpha\cos[(1+j)\lambda-(1+j)\lambda']b_{3/2}^{(j)} \\ &- (1/4)\alpha\cos[(1-j)\lambda+(1-j)\lambda']b_{3/2}^{(j)}.\end{aligned} \tag{298}$$

1.2.6 Simplification of S_D

The above terms can be simplified by taking into account that the index j goes from $-\infty$ to ∞. In general, we want to do variable change so that the multiplier for λ' is j. Since $b_{1/2}^{(-j)} = b_{1/2}^{(j)}$, we can replace

$$\begin{aligned}
\left(S_D^{(j)}\right)_e &= (1/4)\,(2j - \alpha D)\cos((1+j)\lambda - j\lambda' - \varpi)b_{1/2}^{(j)} && (299)\\
&+ (1/4)\,(-2j - \alpha D)\cos((1-j)\lambda + j\lambda' - \varpi)b_{1/2}^{(j)}\\
&\mapsto (1/2)\,(-2j - \alpha D)\cos((1-j)\lambda + j\lambda' - \varpi)b_{1/2}^{(j)}, && (300)\\
\left(S_D^{(j)}\right)_{e^2} &= (1/8)\left(-4j^2 + 2\alpha D + \alpha^2 D^2\right)\cos[j(\lambda - \lambda')]b_{1/2}^{(j)} && (301)\\
&+ (1/16)\left(5j + 4j^2 - 2\alpha D - 4j\alpha D + \alpha^2 D^2\right)\cos[(2+j)\lambda - j\lambda' - 2\varpi]b_{1/2}^{(j)}\\
&+ (1/16)\left(-5j + 4j^2 - 2\alpha D + 4j\alpha D + \alpha^2 D^2\right)\cos[(2-j)\lambda + j\lambda' - 2\varpi]b_{1/2}^{(j)}\\
&\mapsto (1/8)\left(-4j^2 + 2\alpha D + \alpha^2 D^2\right)\cos[j(\lambda - \lambda')]b_{1/2}^{(j)} && (302)\\
&+ (1/8)\left(-5j + 4j^2 - 2\alpha D + 4j\alpha D + \alpha^2 D^2\right)\cos[(2+j)\lambda - j\lambda' - 2\varpi]b_{1/2}^{(j)}
\end{aligned}$$

Consider now the component

$$\begin{aligned}
\left(S_D^{(j)}\right)_{e'} &= (1/4)\,(1 + 2j + \alpha D)\cos[(1+j)\lambda' - j\lambda - \varpi']b_{1/2}^{(j)} && (303)\\
&+ (1/4)\,(1 - 2j + \alpha D)\cos[(1-j)\lambda' + j\lambda - \varpi']b_{1/2}^{(j)}.
\end{aligned}$$

For the first and second rows, substitute $j' = 1 + j$ and $j'' = 1 - j$, respectively, which implies $j = j' - 1 = 1 - j''$. Then,

$$\begin{aligned}
(1+j)\lambda' - j\lambda &= j'\lambda' - (j'-1)\lambda, && (304)\\
(1-j)\lambda' + j\lambda &= j''\lambda' - (j''-1)\lambda, && (305)\\
1 + 2j + \alpha D &= -1 + 2j' + \alpha D, && (306)\\
1 - 2j + \alpha D &= -1 + 2j'' + \alpha D, && (307)
\end{aligned}$$

and we can write

$$\left(S_D^{(j)}\right)_{e'} = (1/2)\,(-1 + 2j + \alpha D)\cos[j\lambda' + (1-j)\lambda - \varpi']b_{1/2}^{(j-1)} \quad (308)$$

Consider now the component

$$\begin{aligned}
\left(S_D^{(j)}\right)_{e'^2} &= (1/8)\left(-4j^2 + 2\alpha D + \alpha^2 D^2\right)\cos[j(\lambda - \lambda')]b_{1/2}^{(j)} && (309)\\
&+ (1/16)\left(4 - 9j + 4j^2 + 6\alpha D - 4j\alpha D + \alpha^2 D^2\right)\cos[(2-j)\lambda' + j\lambda - 2\varpi']b_{1/2}^{(j)}\\
&+ (1/16)\left(4 + 9j + 4j^2 + 6\alpha D + 4j\alpha D + \alpha^2 D^2\right)\cos[(2+j)\lambda' - j\lambda - 2\varpi']b_{1/2}^{(j)}.
\end{aligned}$$

For the second row, substitute $j' = 2 - j$, which implies $j = 2 - j'$ and $j^2 = j'^2 - 4j' + 4$. Then,

$$(2-j)\lambda' + j\lambda - 2\varpi' = j'\lambda' + (2-j')\lambda - 2\varpi' \quad (310)$$

and

$$\begin{aligned} & 4 - 9j + 4j^2 + 6\alpha D - 4j\alpha D + \alpha^2 D^2 && (311)\\ = \;& 4 - 9(2 - j') + 4(j'^2 - 4j' + 4) + 6\alpha D - 4(2 - j')\alpha D + \alpha^2 D^2 \\ = \;& (4 - 18 + 16) + (9 - 16)j' + 4j'^2 + (6 - 8)\alpha D - 4j'\alpha D + \alpha^2 D^2 && (312)\\ = \;& 2 - 7j' + 4j'^2 - 2\alpha D - 4j'\alpha D + \alpha^2 D^2 && (313)\end{aligned}$$

For the third row, substitute $j' = 2 + j$, which implies $j = j' - 2$ and $j^2 = j'^2 - 4j' + 4$. Then,

$$(2 + j)\lambda' - j\lambda - 2\varpi = j'\lambda' + (2 - j')\lambda - 2\varpi' \tag{314}$$

and

$$\begin{aligned} & 4 + 9j + 4j^2 + 6\alpha D + 4j\alpha D + \alpha^2 D^2 && (315)\\ = \;& 4 + 9(j' - 2) + 4(j'^2 - 4j' + 4) + 6\alpha D + 4(j' - 2)\alpha D + \alpha^2 D^2 \\ = \;& (4 - 18 + 16) + (9 - 16)j' + 4j'^2 + (6 - 8)\alpha D + 4j'\alpha D + \alpha^2 D^2 && (316)\\ = \;& 2 - 7j' + 4j'^2 - 2\alpha D + 4j'\alpha D + \alpha^2 D^2. && (317)\end{aligned}$$

Thus, we can write

$$\begin{aligned} \left(S_D^{(j)}\right)_{e'^2} &= (1/8)\left(-4j^2 + 2\alpha D + \alpha^2 D^2\right)\cos[j(\lambda - \lambda')]b_{1/2}^{(j)} && (318)\\ &+ (1/8)\left(2 - 7j + 4j^2 - 2\alpha D + 4j\alpha D + \alpha^2 D^2\right)\cos[j\lambda' + (2 - j)\lambda - 2\varpi']b_{1/2}^{(j-2)}.\end{aligned}$$

Now consider the component

$$\begin{aligned} \left(S_D^{(j)}\right)_{ee'} &= (1/8)\left(2j + 4j^2 - 2\alpha D - \alpha^2 D^2\right)\cos[(1 + j)\lambda - (1 + j)\lambda' - \varpi + \varpi']b_{1/2}^{(j)} && (319)\\ &+ (1/8)\left(-2j + 4j^2 - 2\alpha D - \alpha^2 D^2\right)\cos[(1 - j)\lambda - (1 - j)\lambda' - \varpi + \varpi']b_{1/2}^{(j)} \\ &+ (1/8)\left(2j - 4j^2 - 2\alpha D + 4j\alpha D - \alpha^2 D^2\right)\cos[(1 + j)\lambda + (1 - j)\lambda' - \varpi - \varpi']b_{1/2}^{(j)} \\ &+ (1/8)\left(-2j - 4j^2 - 2\alpha D - 4j\alpha D - \alpha^2 D^2\right)\cos[(1 - j)\lambda + (1 + j)\lambda' - \varpi - \varpi']b_{1/2}^{(j)},\end{aligned}$$

For the first term, substitute $j' = -(1 + j)$, which implies $j = -(1 + j')$ and $j^2 = 1 + 2j' + j'^2$. Then,

$$(1 + j)\lambda - (1 + j)\lambda' = -j'\lambda + j'\lambda' \tag{320}$$

and

$$\begin{aligned} 2j + 4j^2 - 2\alpha D - \alpha^2 D^2 &= 2(-1 - j') + 4(1 + 2j' + j'^2) - 2\alpha D - \alpha^2 D^2 && (321)\\ &= 2 + 6j' + 4j'^2 - 2\alpha D - \alpha^2 D^2. && (322)\end{aligned}$$

For the second term, substitute $j' = -(1 - j)$, which implies $j = j' + 1$ and $j^2 = 1 + 2j' + j'^2$. Then,

$$(1 - j)\lambda - (1 - j)\lambda' = -j'\lambda + j'\lambda' \tag{323}$$

and

$$-2j + 4j^2 - 2\alpha D - \alpha^2 D^2 = -2(1 + j') + 4(1 + 2j' + j'^2) - 2\alpha D - \alpha^2 D^2 \quad (324)$$
$$= 2 + 6j' + 4j'^2 - 2\alpha D - \alpha^2 D^2. \quad (325)$$

For the third term, substitute $j' = 1 - j$, which implies $j = 1 - j'$ and $j^2 = 1 - 2j' + j'^2$. Then,

$$(1 + j)\lambda - (1 - j)\lambda' = (2 - j')\lambda + j'\lambda' \quad (326)$$

and

$$2j - 4j^2 - 2\alpha D + 4j\alpha D - \alpha^2 D^2 = 2(1 - j') - 4(1 - 2j' + j'^2) - 2\alpha D + 4(1 - j')\alpha D - \alpha^2 D^2 \quad (327)$$
$$= (2 - 4) + (-2 + 8)j' - 4j'^2 + (-2 + 4)\alpha D - 4j'\alpha D - \alpha^2 D^2 \quad (328)$$
$$= -2 + 6j' - 4j'^2 + 2\alpha D - 4j'\alpha D - \alpha^2 D^2. \quad (329)$$

For the fourth term, substitute $j' = 1 + j$, which implies $j = j' - 1$ and $j^2 = 1 - 2j' + j'^2$. Then,

$$(1 - j)\lambda + (1 + j)\lambda' = (2 - j')\lambda + j'\lambda' \quad (330)$$

and

$$-2j - 4j^2 - 2\alpha D - 4j\alpha D - \alpha^2 D^2 = -2(j' - 1) - 4(1 - 2j' + j'^2) - 2\alpha D - 4(j' - 1)\alpha D - \alpha^2 D^2 \quad (331)$$
$$= (2 - 4) + (-2 + 8)j' - 4j'^2 + (-2 + 4)\alpha D - 4j'\alpha D - \alpha^2 D^2 \quad (332)$$
$$= -2 + 6j' - 4j'^2 + 2\alpha D - 4j'\alpha D - \alpha^2 D^2. \quad (333)$$

Thus, we can write

$$\left(S_D^{(j)}\right)_{ee'} = (1/4)\left(2 + 6j + 4j^2 - 2\alpha D - \alpha^2 D^2\right)\cos[j\lambda' - j\lambda - \varpi + \varpi']b_{1/2}^{j+1} \quad (334)$$
$$+ (1/4)\left(-2 + 6j - 4j^2 + 2\alpha D - 4j\alpha D - \alpha^2 D^2\right)\cos[j\lambda' + (2 - j)\lambda - \varpi - \varpi']b_{1/2}^{(j-1)}.$$

Now consider the remaining terms

$$\left(S_D^{(j)}\right)_{ss'} = (1/2)\alpha\cos[(1 - j)\lambda - (1 - j)\lambda' - \Omega + \Omega']b_{3/2}^{(j)} \quad (335)$$
$$+ (1/2)\alpha\cos[(1 + j)\lambda - (1 + j)\lambda' - \Omega + \Omega']b_{3/2}^{(j)}$$
$$- (1/2)\alpha\cos[(1 - j)\lambda + (1 + j)\lambda' - \Omega - \Omega']b_{3/2}^{(j)}$$
$$- (1/2)\alpha\cos[(1 + j)\lambda + (1 - j)\lambda' - \Omega - \Omega']b_{3/2}^{(j)},$$
$$\left(S_D^{(j)}\right)_{s^2} = (1/4)\alpha\cos[(1 + j)\lambda + (1 - j)\lambda' - 2\Omega]b_{3/2}^{(j)} \quad (336)$$
$$+ (1/4)\alpha\cos[(1 - j)\lambda + (1 + j)\lambda' - 2\Omega]b_{3/2}^{(j)}$$
$$- (1/4)\alpha\cos[(1 + j)\lambda - (1 + j)\lambda']b_{3/2}^{(j)}$$
$$- (1/4)\alpha\cos[(1 - j)\lambda + (1 - j)\lambda']b_{3/2}^{(j)},$$
$$\left(S_D^{(j)}\right)_{s'^2} = (1/4)\alpha\cos[(1 + j)\lambda' + (1 - j)\lambda - 2\Omega']b_{3/2}^{(j)} \quad (337)$$
$$+ (1/4)\alpha\cos[(1 - j)\lambda' + (1 + j)\lambda - 2\Omega']b_{3/2}^{(j)}$$
$$- (1/4)\alpha\cos[(1 + j)\lambda - (1 + j)\lambda']b_{3/2}^{(j)}$$
$$- (1/4)\alpha\cos[(1 - j)\lambda + (1 - j)\lambda']b_{3/2}^{(j)}.$$

First, let's investigate an expression of the form

$$\cos[(1+j)\lambda-(1+j)\lambda'+\delta]b_{3/2}^{(j)}+\cos[(1-j)\lambda-(1-j)\lambda'+\delta]b_{3/2}^{(j)} \tag{338}$$

and make substitutions $j'=-(1+j)$ and $j''=-(1-j)$, which imply $j=-(1+j')=j''+1$ to get

$$\cos[j'\lambda'-j'\lambda+\delta]b_{3/2}^{(1+j')}+\cos[j''\lambda-j''\lambda'+\delta]b_{3/2}^{(1+j'')}. \tag{339}$$

Similarly, for expressions of the form

$$\cos[(1-j)\lambda+(1+j)\lambda'+\delta]b_{3/2}^{(j)}+\cos[(1+j)\lambda+(1-j)\lambda'+\delta]b_{3/2}^{(j)} \tag{340}$$

and make substitutions $j'=1+j$ and $j''=1-j$, which imply $j=j'-1=1-j''$ to get

$$\cos[(2-j')\lambda+j'\lambda'+\delta]b_{3/2}^{(j'-1)}+\cos[(2-j'')\lambda+j''\lambda'+\delta]b_{3/2}^{(j''-1)}. \tag{341}$$

Collecting the terms, we finally get

$$\left(S_D^{(j)}\right)_1 = \frac{1}{2}\cos[j(\lambda-\lambda')]b_{1/2}^{(j)}, \tag{342}$$

$$\left(S_D^{(j)}\right)_e = (1/2)\,(-2j-\alpha D)\cos(j\lambda'+(1-j)\lambda-\varpi)b_{1/2}^{(j)}, \tag{343}$$

$$\left(S_D^{(j)}\right)_{e'} = (1/2)\,(-1+2j+\alpha D)\cos[j\lambda'+(1-j)\lambda-\varpi']b_{1/2}^{(j-1)} \tag{344}$$

$$\begin{aligned}\left(S_D^{(j)}\right)_{e^2} &= (1/8)\left(-4j^2+2\alpha D+\alpha^2D^2\right)\cos[j(\lambda-\lambda')]b_{1/2}^{(j)} \\ &+ (1/8)\left(-5j+4j^2-2\alpha D+4j\alpha D+\alpha^2D^2\right)\cos[j\lambda'+(2-j)\lambda-2\varpi]b_{1/2}^{(j)},\end{aligned} \tag{345}$$

$$\begin{aligned}\left(S_D^{(j)}\right)_{e'^2} &= (1/8)\left(-4j^2+2\alpha D+\alpha^2D^2\right)\cos[j(\lambda-\lambda')]b_{1/2}^{(j)} \\ &+ (1/8)\left(2-7j+4j^2-2\alpha D+4j\alpha D+\alpha^2D^2\right)\cos[j\lambda'+(2-j)\lambda-2\varpi']b_{1/2}^{(j-2)},\end{aligned} \tag{346}$$

$$\begin{aligned}\left(S_D^{(j)}\right)_{ee'} &= (1/4)\left(2+6j+4j^2-2\alpha D-\alpha^2D^2\right)\cos[j\lambda'-j\lambda-\varpi+\varpi']b_{1/2}^{j+1} \\ &+ (1/4)\left(-2+6j-4j^2+2\alpha D-4j\alpha D-\alpha^2D^2\right)\cos[j\lambda'+(2-j)\lambda-\varpi-\varpi']b_{1/2}^{(j-1)},\end{aligned} \tag{347}$$

$$\begin{aligned}\left(S_D^{(j)}\right)_{ss'} &= \alpha\cos[j\lambda'-j\lambda-\Omega+\Omega']b_{3/2}^{(j+1)} \\ &- \alpha\cos[j\lambda'+(2-j)\lambda-\Omega-\Omega']b_{3/2}^{(j-1)},\end{aligned} \tag{348}$$

$$\begin{aligned}\left(S_D^{(j)}\right)_{s^2} &= -(1/2)\alpha\cos[j(\lambda-\lambda')]b_{3/2}^{(j+1)} \\ &+ (1/2)\alpha\cos[j\lambda'+(2-j)\lambda-2\Omega]b_{3/2}^{(j-1)},\end{aligned} \tag{349}$$

$$\begin{aligned}\left(S_D^{(j)}\right)_{s'^2} &= -(1/2)\alpha\cos[j(\lambda-\lambda')]b_{3/2}^{(j+1)} \\ &+ (1/2)\alpha\cos[j\lambda'+(2-j)\lambda-2\Omega']b_{3/2}^{(j-1)}.\end{aligned} \tag{350}$$

1.3 Expansion of S_E and S_I

We will expand the expression (20)

$$S_E = \left[-\left(\frac{r}{a}\right)\left(\frac{a'}{r'}\right)^2\right]\cos\psi \tag{351}$$

where

$$
\begin{aligned}
(\cos\psi)_1 &= \cos(\lambda-\lambda'), && (352)\\
(\cos\psi)_e &= \cos(2\lambda-\lambda'-\varpi)-\cos(\lambda'-\varpi), && (353)\\
(\cos\psi)_{e'} &= \cos(2\lambda'-\lambda-\varpi')-\cos(\lambda-\varpi'), && (354)\\
(\cos\psi)_{ee'} &= \cos(2\lambda-2\lambda'-\varpi+\varpi')-\cos(2\lambda-\varpi-\varpi'), && (355)\\
&- \cos(2\lambda'-\varpi'-\varpi)+\cos(\varpi-\varpi'),\\
(\cos\psi)_{e^2} &= \frac{9}{8}\cos(3\lambda-\lambda'-2\varpi)-\cos(\lambda-\lambda')-\frac{1}{8}\cos(\lambda+\lambda'-2\varpi), && (356)\\
(\cos\psi)_{e'^2} &= \frac{9}{8}\cos(3\lambda'-\lambda-2\varpi')-\cos(\lambda-\lambda')-\frac{1}{8}\cos(\lambda'+\lambda-2\varpi'), && (357)\\
(\cos\psi)_{ss'} &= 2\cos(\lambda-\lambda'-\Omega+\Omega')-2\cos(\lambda+\lambda'-\Omega-\Omega'), && (358)\\
(\cos\psi)_{s^2} &= \cos(\lambda+\lambda'-2\Omega)-\cos(\lambda-\lambda'), && (359)\\
(\cos\psi)_{s'^2} &= \cos(\lambda'+\lambda-2\Omega')-\cos(\lambda-\lambda'). && (360)
\end{aligned}
$$

We also use

$$
\begin{aligned}
\frac{r}{a} &= 1+\zeta = 1-e\cos(\lambda-\varpi)+\frac{1}{2}e^2[1-\cos(2\lambda-2\varpi)]+\mathcal{O}(e^3), && (361)\\
\left(\frac{a'}{r'}\right)^2 &= 1-2\zeta'+3\zeta'^2+\mathcal{O}(\zeta'^3) && (362)\\
&= 1+2e'\cos(\lambda'-\varpi')+\frac{1}{2}e'^2+\frac{5}{2}e'^2\cos(2\lambda'-2\varpi')+\mathcal{O}(e'^3) && (363)
\end{aligned}
$$

to obtain for the bracket part of (351)

$$
\begin{aligned}
[\,]_1 &= -1, && (364)\\
[\,]_e &= \cos(\lambda-\varpi), && (365)\\
[\,]_{e'} &= -2\cos(\lambda'-\varpi'), && (366)\\
[\,]_{ee'} &= 2\cos(\lambda-\varpi)\cos(\lambda'-\varpi') && (367)\\
&= \cos(\lambda+\lambda'-\varpi-\varpi')+\cos(\lambda-\lambda'-\varpi+\varpi'),\\
[\,]_{e^2} &= -\frac{1}{2}[1-\cos(2\lambda-2\varpi)], && (368)\\
[\,]_{e'^2} &= -\frac{1}{2}[1+5\cos(2\lambda'-2\varpi')]. && (369)
\end{aligned}
$$

Thus, we can compute the different components of $\mathcal{S}_E$ by straight-forward multiplication

$$
\begin{aligned}
(\mathcal{S}_E)_1 &= [\,]_1 (\cos\psi)_1 && (370)\\
&= -\cos(\lambda-\lambda') && (371)\\
(\mathcal{S}_E)_e &= [\,]_e (\cos\psi)_1 + [\,]_1 (\cos\psi)_e && (372)\\
&= \cos(\lambda-\varpi)\cos(\lambda-\lambda') - \cos(2\lambda-\lambda'-\varpi) + \cos(\lambda'-\varpi), && (373)\\
&= \frac{1}{2}\cos(2\lambda-\lambda'-\varpi) + \frac{1}{2}\cos(\lambda'-\varpi) - \cos(2\lambda-\lambda'-\varpi) + \cos(\lambda-\varpi) && (374)\\
&= -\frac{1}{2}\cos(2\lambda-\lambda'-\varpi) + \frac{3}{2}\cos(\lambda'-\varpi), && (375)\\
(\mathcal{S}_E)_{e'} &= [\,]_{e'} (\cos\psi)_1 + [\,]_1 (\cos\psi)_{e'} && (376)\\
&= -2\cos(\lambda'-\varpi')\cos(\lambda-\lambda') - \cos(2\lambda'-\lambda-\varpi') + \cos(\lambda-\varpi') && (377)\\
&= -\cos(\lambda-\varpi') - \cos(2\lambda'-\lambda-\varpi') - \cos(2\lambda'-\lambda-\varpi') + \cos(\lambda-\varpi') && (378)\\
&= -2\cos(2\lambda'-\lambda-\varpi') && (379)\\
&= -2\cos(\lambda-2\lambda'+\varpi'), && (380)\\
(\mathcal{S}_E)_{ss'} &= [\,]_1 (\cos\psi)_{ss'} && (381)\\
&= -2\cos(\lambda'-\lambda-\Omega'+\Omega) + 2\cos(\lambda'+\lambda-\Omega'-\Omega) && (382)\\
(\mathcal{S}_E)_{s^2} &= [\,]_1 (\cos\psi)_{s^2} && (383)\\
&= -\cos(\lambda+\lambda'-2\Omega) + \cos(\lambda-\lambda') && (384)\\
(\mathcal{S}_E)_{s'^2} &= [\,]_1 (\cos\psi)_{s'^2} && (385)\\
&= -\cos(\lambda+\lambda'-2\Omega') + \cos(\lambda-\lambda') && (386)
\end{aligned}
$$

For the more complex components, it makes sense to compute the terms separately. For

$$
(\mathcal{S}_E)_{e^2} = [\,]_{e^2} (\cos\psi)_1 + [\,]_1 (\cos\psi)_{e^2} + [\,]_e (\cos\psi)_e\,, \qquad (387)
$$

we compute

$$
\begin{aligned}
[\,]_{e^2} (\cos\psi)_1 &= -\frac{1}{2}[1-\cos(2\lambda-2\varpi)]\cos(\lambda-\lambda') && (388)\\
&= -\frac{1}{2}\cos(\lambda-\lambda') + \frac{1}{4}\cos(3\lambda-\lambda'-2\varpi) + \frac{1}{4}\cos(\lambda+\lambda'-2\varpi), && (389)\\
[\,]_1 (\cos\psi)_{e^2} &= -\frac{9}{8}\cos(3\lambda-\lambda'-2\varpi) + \cos(\lambda-\lambda') + \frac{1}{8}\cos(\lambda+\lambda'-2\varpi), && (390)\\
[\,]_e (\cos\psi)_e &= \cos(\lambda-\varpi)\left[\cos(2\lambda-\lambda'-\varpi) - \cos(\lambda'-\varpi)\right] && (391)\\
&= \frac{1}{2}\cos(3\lambda-\lambda'-2\varpi) + \frac{1}{2}\cos(\lambda-\lambda') - \frac{1}{2}\cos(\lambda+\lambda'-2\varpi) - \frac{1}{2}\cos(\lambda-\lambda') && (392)\\
&= \frac{1}{2}\cos(3\lambda-\lambda'-2\varpi) - \frac{1}{2}\cos(\lambda+\lambda'-2\varpi) && (393)
\end{aligned}
$$

to obtain

$$
(\mathcal{S}_E)_{e^2} = \frac{1}{2}\cos(\lambda-\lambda') - \frac{3}{8}\cos(3\lambda-\lambda'-2\varpi) - \frac{1}{8}\cos(\lambda+\lambda'-2\varpi). \qquad (394)
$$

For

$$(\mathcal{S}_E)_{e'^2} = [\,]_{e'^2}(\cos\psi)_1 + [\,]_1(\cos\psi)_{e'^2} + [\,]_{e'}(\cos\psi)_{e'}, \tag{395}$$

we compute

$$\begin{aligned}
[\,]_{e'^2}(\cos\psi)_1 &= -\frac{1}{2}[1+5\cos(2\lambda'-2\varpi')]\cos(\lambda-\lambda') && (396)\\
&= -\frac{1}{2}\cos(\lambda-\lambda') - \frac{5}{4}\cos(3\lambda'-\lambda-2\varpi') - \frac{5}{4}\cos(\lambda'+\lambda-2\varpi'), && (397)\\
[\,]_1(\cos\psi)_{e'^2} &= -\frac{9}{8}\cos(3\lambda'-\lambda-2\varpi') + \cos(\lambda-\lambda') + \frac{1}{8}\cos(\lambda'+\lambda-2\varpi'), && (398)\\
[\,]_{e'}(\cos\psi)_{e'} &= -2\cos(\lambda'-\varpi')\left[\cos(2\lambda'-\lambda-\varpi') - \cos(\lambda-\varpi')\right] && (399)\\
&= -\cos(3\lambda'-\lambda-2\varpi') - \cos(\lambda'-\lambda) + \cos(\lambda'+\lambda-2\varpi') + \cos(\lambda'-\lambda) && (400)\\
&= -\cos(3\lambda'-\lambda-2\varpi') + \cos(\lambda'+\lambda-2\varpi') && (401)
\end{aligned}$$

to obtain

$$(\mathcal{S}_E)_{e'^2} = \frac{1}{2}\cos(\lambda-\lambda') - \frac{1}{8}\cos(\lambda'+\lambda-2\varpi') - \frac{27}{8}\cos(3\lambda'-\lambda-2\varpi'). \tag{402}$$

For

$$(\mathcal{S}_E)_{ee'} = [\,]_e(\cos\psi)_{e'} + [\,]_{e'}(\cos\psi)_e + [\,]_1(\cos\psi)_{ee'} + [\,]_{ee'}(\cos\psi)_1 \tag{403}$$

we compute

$$\begin{aligned}
[\,]_e(\cos\psi)_{e'} &= \cos(\lambda-\varpi)\left[\cos(2\lambda'-\lambda-\varpi') - \cos(\lambda-\varpi')\right] && (404)\\
&= \frac{1}{2}\cos(2\lambda'-\varpi-\varpi') + \frac{1}{2}\cos(2\lambda-2\lambda'-\varpi+\varpi') && (405)\\
&- \frac{1}{2}\cos(2\lambda-\varpi-\varpi') - \frac{1}{2}\cos(\varpi-\varpi'),\\
[\,]_{e'}(\cos\psi)_e &= -2\cos(\lambda'-\varpi')\left[\cos(2\lambda-\lambda'-\varpi) - \cos(\lambda'-\varpi)\right] && (406)\\
&= -\cos(2\lambda-\varpi-\varpi') - \cos(2\lambda'-2\lambda+\varpi-\varpi') && (407)\\
&+ \cos(2\lambda'-\varpi-\varpi') + \cos(\varpi-\varpi'),\\
[\,]_{ee'}(\cos\psi)_1 &= \left[\cos(\lambda+\lambda'-\varpi-\varpi') + \cos(\lambda-\lambda'-\varpi+\varpi')\right]\cos(\lambda-\lambda') && (408)\\
&= \frac{1}{2}\cos(2\lambda-\varpi-\varpi') + \frac{1}{2}\cos(2\lambda'-\varpi-\varpi') && (409)\\
&+ \frac{1}{2}\cos(2\lambda-2\lambda'-\varpi+\varpi') + \frac{1}{2}\cos(\varpi-\varpi'),\\
[\,]_1(\cos\psi)_{ee'} &= -\cos(2\lambda-2\lambda'-\varpi+\varpi') + \cos(2\lambda-\varpi-\varpi'), && (410)\\
&+ \cos(2\lambda'-\varpi'-\varpi) - \cos(\varpi-\varpi').
\end{aligned}$$

to obtain

$$(\mathcal{S}_E)_{ee'} = 3\cos(2\lambda'-\varpi-\varpi') - \cos(2\lambda-2\lambda'-\varpi+\varpi'). \tag{411}$$

We will expand the expression (21)

$$S_I = \left[-\left(\frac{r'}{a'}\right)\left(\frac{a}{r}\right)^2\right]\cos\psi. \tag{412}$$

We again use

$$\frac{r'}{a'} = 1+\zeta' = 1-e'\cos(\lambda'-\varpi')+\frac{1}{2}e'^2[1-\cos(2\lambda'-2\varpi')]+\mathcal{O}(e'^3), \tag{413}$$

$$\left(\frac{a}{r}\right)^2 = 1-2\zeta+3\zeta^2+\mathcal{O}(\zeta^3) \tag{414}$$

$$= 1+2e\cos(\lambda-\varpi)+\frac{1}{2}e^2+\frac{5}{2}e^2\cos(2\lambda-2\varpi)+\mathcal{O}(e^3) \tag{415}$$

to obtain for the bracket part of (412)

$$[\,]_1 = -1, \tag{416}$$

$$[\,]_{e'} = \cos(\lambda'-\varpi'), \tag{417}$$

$$[\,]_e = -2\cos(\lambda-\varpi), \tag{418}$$

$$[\,]_{ee'} = \cos(\lambda'+\lambda-\varpi'-\varpi)+\cos(\lambda'-\lambda-\varpi'+\varpi), \tag{419}$$

$$[\,]_{e'^2} = -\frac{1}{2}[1-\cos(2\lambda'-2\varpi')], \tag{420}$$

$$[\,]_{e^2} = -\frac{1}{2}[1+5\cos(2\lambda-2\varpi)]. \tag{421}$$

It's important not to confuse the brackets for S_I with brackets for S_E. Again by straight-forward multiplication

$$(S_I)_1 = [\,]_1(\cos\psi)_1 \tag{422}$$

$$= -\cos(\lambda-\lambda') \tag{423}$$

$$(S_I)_e = [\,]_e(\cos\psi)_1+[\,]_1(\cos\psi)_e \tag{424}$$

$$= -2\cos(\lambda-\varpi)\cos(\lambda-\lambda')-\cos(2\lambda-\lambda'-\varpi)+\cos(\lambda'-\varpi) \tag{425}$$

$$= -\cos(2\lambda-\lambda'-\varpi)-\cos(\lambda'-\varpi)-\cos(2\lambda-\lambda'-\varpi)+\cos(\lambda'-\varpi) \tag{426}$$

$$= -2\cos(2\lambda-\lambda'-\varpi) \tag{427}$$

$$= -2\cos(\lambda'-2\lambda+\varpi) \tag{428}$$

$$(S_I)_{e'} = [\,]_{e'}(\cos\psi)_1+[\,]_1(\cos\psi)_{e'} \tag{429}$$

$$= \cos(\lambda'-\varpi')\cos(\lambda'-\lambda)-\cos(2\lambda'-\lambda-\varpi')+\cos(\lambda-\varpi') \tag{430}$$

$$= \frac{1}{2}\cos(2\lambda'-\lambda-\varpi')+\frac{1}{2}\cos(\lambda-\varpi')-\cos(2\lambda'-\lambda-\varpi')+\cos(\lambda-\varpi') \tag{431}$$

$$= -\frac{1}{2}\cos(2\lambda'-\lambda-\varpi')+\frac{3}{2}\cos(\lambda-\varpi') \tag{432}$$

$$(S_I)_{ss'} = [\,]_1(\cos\psi)_{ss'} \tag{433}$$

$$= -2\cos(\lambda'-\lambda-\Omega'+\Omega)+2\cos(\lambda'+\lambda-\Omega'-\Omega) \tag{434}$$

$$(S_I)_{s^2} = [\,]_1(\cos\psi)_{s^2} \tag{435}$$

$$= -\cos(\lambda+\lambda'-2\Omega)+\cos(\lambda-\lambda') \tag{436}$$

$$(S_I)_{s'^2} = [\,]_1(\cos\psi)_{s'^2} \tag{437}$$

$$= -\cos(\lambda+\lambda'-2\Omega')+\cos(\lambda-\lambda') \tag{438}$$

For

$$\left(\mathcal{S}_I\right)_{e^2} = [\,]_1 (\cos\psi)_{e^2} + [\,]_{e^2} (\cos\psi)_1 + [\,]_e (\cos\psi)_e . \tag{439}$$

we compute

$$\begin{aligned}
[\,]_1 (\cos\psi)_{e^2} &= -\frac{9}{8}\cos(3\lambda - \lambda' - 2\varpi) + \cos(\lambda - \lambda') + \frac{1}{8}\cos(\lambda + \lambda' - 2\varpi), && (440)\\
[\,]_{e^2} (\cos\psi)_1 &= -\frac{1}{2}[1 + 5\cos(2\lambda - 2\varpi)]\cos(\lambda - \lambda') && (441)\\
&= -\frac{1}{2}\cos(\lambda - \lambda') - \frac{5}{4}\cos(3\lambda - \lambda' - 2\varpi) + \frac{5}{4}\cos(\lambda + \lambda' - 2\varpi), && (442)\\
[\,]_e (\cos\psi)_e &= -2\cos(\lambda - \varpi)\left[\cos(2\lambda - \lambda' - \varpi) - \cos(\lambda' - \varpi)\right] && (443)\\
&= -\cos(3\lambda - \lambda' - 2\varpi) - \cos(\lambda - \lambda') + \cos(\lambda' + \lambda - 2\varpi) + \cos(\lambda' - \lambda) && (444)\\
&= -\cos(3\lambda - \lambda' - 2\varpi) + \cos(\lambda' + \lambda - 2\varpi). && (445)
\end{aligned}$$

to obtain

$$\left(\mathcal{S}_I\right)_{e^2} = \frac{1}{2}\cos(\lambda - \lambda') - \frac{27}{8}\cos(3\lambda - \lambda' - 2\varpi) - \frac{1}{8}\cos(\lambda + \lambda' - 2\varpi). \tag{446}$$

For

$$\left(\mathcal{S}_I\right)_{e'^2} = [\,]_1 (\cos\psi)_{e'^2} + [\,]_{e'^2} (\cos\psi)_1 + [\,]_{e'} (\cos\psi)_{e'} . \tag{447}$$

we compute

$$\begin{aligned}
[\,]_1 (\cos\psi)_{e'^2} &= -\frac{9}{8}\cos(3\lambda' - \lambda - 2\varpi') + \cos(\lambda - \lambda') + \frac{1}{8}\cos(\lambda' + \lambda - 2\varpi'), && (448)\\
[\,]_{e'^2} (\cos\psi)_1 &= -\frac{1}{2}[1 - \cos(2\lambda' - 2\varpi')]\cos(\lambda - \lambda') && (449)\\
&= -\frac{1}{2}\cos(\lambda - \lambda') + \frac{1}{4}\cos(\lambda' - \lambda - 2\varpi') + \frac{1}{4}\cos(3\lambda' + \lambda - 2\varpi') && (450)\\
[\,]_{e'} (\cos\psi)_{e'} &= \cos(\lambda' - \varpi')\left[\cos(2\lambda' - \lambda - \varpi') - \cos(\lambda - \varpi')\right] && (451)\\
&= \frac{1}{2}\cos(3\lambda' - \lambda - 2\varpi') + \frac{1}{2}\cos(\lambda' - \lambda) - \frac{1}{2}\cos(\lambda' + \lambda - 2\varpi') - \frac{1}{2}\cos(\lambda' - \lambda) && (452)\\
&= \frac{1}{2}\cos(3\lambda' - \lambda - 2\varpi') - \frac{1}{2}\cos(\lambda' + \lambda - 2\varpi') && (453)
\end{aligned}$$

to obtain

$$\left(\mathcal{S}_I\right)_{e'^2} = \frac{1}{2}\cos(\lambda - \lambda') - \frac{1}{8}\cos(\lambda + \lambda' - 2\varpi') - \frac{3}{8}\cos(3\lambda' - \lambda' - 2\varpi'). \tag{454}$$

For

$$\left(\mathcal{S}_I\right)_{ee'} = [\,]_1 (\cos\psi)_{ee'} + [\,]_{ee'} (\cos\psi)_1 + [\,]_e (\cos\psi)_{e'} . + [\,]_{e'} (\cos\psi)_e \tag{455}$$

we compute

$$
\begin{aligned}
[\,]_1\,(\cos\psi)_{ee'} &= -\cos(2\lambda - 2\lambda' - \varpi + \varpi') + \cos(2\lambda - \varpi - \varpi'), && (456)\\
[\,]_{ee'}\,(\cos\psi)_1 &= \left[\cos(\lambda' + \lambda - \varpi' - \varpi) + \cos(\lambda' - \lambda - \varpi' + \varpi)\right]\cos(\lambda - \lambda') && (457)\\
&= \frac{1}{2}\cos(2\lambda - \varpi' - \varpi) + \frac{1}{2}\cos(2\lambda' - \varpi' - \varpi) && (458)\\
&+ \frac{1}{2}\cos(\varpi' - \varpi) + \frac{1}{2}\cos(2\lambda' - 2\lambda - \varpi' + \varpi)\\
[\,]_e\,(\cos\psi)_{e'} &= -2\cos(\lambda - \varpi)\left[\cos(2\lambda' - \lambda - \varpi') - \cos(\lambda - \varpi')\right] && (459)\\
&= -\cos(2\lambda' - \varpi - \varpi') - \cos(2\lambda - 2\lambda' - \varpi + \varpi') && (460)\\
&+ \cos(2\lambda - \varpi - \varpi') + \cos(\varpi - \varpi')\\
[\,]_{e'}\,(\cos\psi)_e &= \cos(\lambda' - \varpi')\left[\cos(2\lambda - \lambda' - \varpi) - \cos(\lambda' - \varpi)\right] && (461)\\
&= \frac{1}{2}\cos(2\lambda - \varpi - \varpi') + \frac{1}{2}\cos(2\lambda - 2\lambda' - \varpi + \varpi') && (462)\\
&- \frac{1}{2}\cos(2\lambda' - \varpi - \varpi') - \frac{1}{2}\cos(\varpi - \varpi')
\end{aligned}
$$

to obtain

$$
\left(\mathcal{S}_I\right)_{ee'} = -\cos(2\lambda - 2\lambda' - \varpi + \varpi') + 3\cos(2\lambda - \varpi - \varpi'). \quad (463)
$$

Thus, collecting the terms, we obtain

$$
\begin{aligned}
(\mathcal{S}_E)_1 &= -\cos(\lambda - \lambda'), && (464)\\
(\mathcal{S}_E)_e &= -\frac{1}{2}\cos(2\lambda - \lambda' - \varpi) + \frac{3}{2}\cos(\lambda' - \varpi), && (465)\\
(\mathcal{S}_E)_{e'} &= -2\cos(\lambda - 2\lambda' + \varpi'), && (466)\\
(\mathcal{S}_E)_{ss'} &= -2\cos(\lambda' - \lambda - \Omega' + \Omega) + 2\cos(\lambda' + \lambda - \Omega' - \Omega), && (467)\\
(\mathcal{S}_E)_{s^2} &= -\cos(\lambda + \lambda' - 2\Omega) + \cos(\lambda - \lambda'), && (468)\\
(\mathcal{S}_E)_{s'^2} &= -\cos(\lambda + \lambda' - 2\Omega') + \cos(\lambda - \lambda'), && (469)\\
(\mathcal{S}_E)_{e^2} &= \frac{1}{2}\cos(\lambda - \lambda') - \frac{3}{8}\cos(3\lambda - \lambda' - 2\varpi) - \frac{1}{8}\cos(\lambda + \lambda' - 2\varpi), && (470)\\
(\mathcal{S}_E)_{e'^2} &= \frac{1}{2}\cos(\lambda - \lambda') - \frac{1}{8}\cos(\lambda' + \lambda - 2\varpi') - \frac{27}{8}\cos(3\lambda' - \lambda - 2\varpi'), && (471)\\
(\mathcal{S}_E)_{ee'} &= 3\cos(2\lambda' - \varpi - \varpi') - \cos(2\lambda - 2\lambda' - \varpi + \varpi'). && (472)
\end{aligned}
$$

and

$$
\begin{aligned}
(\mathcal{S}_I)_1 &= -\cos(\lambda-\lambda'), && (473)\\
(\mathcal{S}_I)_e &= -2\cos(\lambda'-2\lambda+\varpi), && (474)\\
(\mathcal{S}_I)_{e'} &= -\frac{1}{2}\cos(2\lambda'-\lambda-\varpi')+\frac{3}{2}\cos(\lambda-\varpi'), && (475)\\
(\mathcal{S}_I)_{ss'} &= -2\cos(\lambda'-\lambda-\Omega'+\Omega)+2\cos(\lambda'+\lambda-\Omega'-\Omega), && (476)\\
(\mathcal{S}_I)_{s^2} &= -\cos(\lambda+\lambda'-2\Omega)+\cos(\lambda-\lambda'), && (477)\\
(\mathcal{S}_I)_{s'^2} &= -\cos(\lambda+\lambda'-2\Omega')+\cos(\lambda-\lambda'), && (478)\\
\left(\mathcal{S}_I\right)_{e^2} &= \frac{1}{2}\cos(\lambda-\lambda')-\frac{27}{8}\cos(3\lambda-\lambda'-2\varpi)-\frac{1}{8}\cos(\lambda+\lambda'-2\varpi), && (479)\\
\left(\mathcal{S}_I\right)_{e'^2} &= \frac{1}{2}\cos(\lambda-\lambda')-\frac{1}{8}\cos(\lambda+\lambda'-2\varpi')-\frac{3}{8}\cos(3\lambda'-\lambda'-2\varpi'), && (480)\\
\left(\mathcal{S}_I\right)_{ee'} &= -\cos(2\lambda-2\lambda'-\varpi+\varpi')+3\cos(2\lambda-\varpi-\varpi'). && (481)
\end{aligned}
$$

1.4 Approximation of Periodic Perturbations

Periodic terms in the disturbing function are the terms depending on mean longitude while terms independent of mean longitude are called **secular terms**. Periodic terms are approximated $\mathcal{O}(e)$ while secular terms are approximated $\mathcal{O}(e^2)$. Consequently, the following terms of $\mathcal{S}_D$ are included without simplification

$$
\begin{aligned}
\left(\mathcal{S}_D\right)_1 &= \frac{1}{2}\sum_{j=-\infty}^{\infty}\cos[j(\lambda-\lambda')]b_{1/2}^{(j)}, && (482)\\
\left(\mathcal{S}_D\right)_e &= \frac{1}{2}\sum_{j=-\infty}^{\infty}(-2j-\alpha D)\cos[j\lambda'+(1-j)\lambda-\varpi]b_{1/2}^{(j)}, && (483)\\
\left(\mathcal{S}_D\right)_{e'} &= \frac{1}{2}\sum_{j=-\infty}^{\infty}(-1+2j+\alpha D)\cos[j\lambda'+(1-j)\lambda-\varpi']b_{1/2}^{(j-1)}, && (484)
\end{aligned}
$$

On the other hand, the second-order terms are simplified significantly

$$
\begin{aligned}
\left(\mathcal{S}_D\right)_{e^2} &\approx \frac{1}{8}(2\alpha D+\alpha^2D^2)b_{1/2}^{(j)}, && (485)\\
\left(\mathcal{S}_D\right)_{e'^2} &\approx \frac{1}{8}(2\alpha D+\alpha^2D^2)b_{1/2}^{(j)}, && (486)\\
\left(\mathcal{S}_D\right)_{ee'} &\approx \frac{1}{4}(2-2\alpha D-\alpha^2D^2)b_{1/2}^{(j+1)}\cos(\varpi-\varpi'), && (487)\\
\left(\mathcal{S}_D\right)_{ss'} &\approx \alpha b_{3/2}^{(j+1)}\cos(\Omega-\Omega'), && (488)\\
\left(\mathcal{S}_D\right)_{s^2} &\approx -\frac{1}{2}\alpha b_{3/2}^{(j+1)}, && (489)\\
\left(\mathcal{S}_D\right)_{s'^2} &\approx -\frac{1}{2}\alpha b_{3/2}^{(j+1)}. && (490)
\end{aligned}
$$

Thus, we obtain the approximation

$$\begin{aligned}
\mathcal{S}_D &\approx \frac{1}{2}\sum_{j=-\infty}^{\infty}\cos[j(\lambda-\lambda')]b_{1/2}^{(j)} \\
&+ \frac{e}{2}\sum_{j=-\infty}^{\infty}(-2j-\alpha D)\cos[j\lambda'+(1-j)\lambda-\varpi]b_{1/2}^{(j)} \\
&+ \frac{e'}{2}\sum_{j=-\infty}^{\infty}(-1+2j+\alpha D)\cos[j\lambda'+(1-j)\lambda-\varpi']b_{1/2}^{(j-1)} \\
&+ \frac{1}{8}(e^2+e'^2)(2\alpha D+\alpha^2D^2)b_{1/2}^{(j)} \\
&+ \frac{ee'}{4}(2-2\alpha D-\alpha^2D^2)b_{1/2}^{(j+1)}\cos(\varpi-\varpi') \\
&+ ss'\alpha b_{3/2}^{(j+1)}\cos(\Omega-\Omega'), \\
&- \frac{1}{2}\alpha(s^2+s'^2)b_{3/2}^{(j+1)}.
\end{aligned} \tag{491}$$

The expressions for $\mathcal{S}_E$ and $\mathcal{S}_I$ simplify even more

$$\mathcal{S}_E = -\cos(\lambda-\lambda') - \frac{1}{2}e\cos(2\lambda-\lambda'-\varpi) + \frac{3}{2}e\cos(\lambda'-\varpi) - 2e'\cos(\lambda-2\lambda'+\varpi'), \tag{492}$$

$$\mathcal{S}_I = -\cos(\lambda-\lambda') - 2e\cos(\lambda'-2\lambda+\varpi) - \frac{1}{2}e'\cos(2\lambda'-\lambda-\varpi') + \frac{3}{2}e'\cos(\lambda-\varpi'). \tag{493}$$

Now we can finally write approximations for the disturbing function of the first body

$$\mathcal{S} := \mathcal{S}_0 + \mathcal{S}_1 + \mathcal{S}_2 = \mathcal{S}_D + \alpha\mathcal{S}_{\mathcal{E}}, \tag{494}$$

where we expand by order

$$\mathcal{S}_0 = \frac{1}{2}\sum_{j=-\infty}^{\infty}b_{1/2}^{(j)}\cos[j(\lambda-\lambda')] - \alpha\cos(\lambda-\lambda'), \tag{495}$$

$$\begin{aligned}
\mathcal{S}_1 &= \frac{e}{2}\sum_{j=-\infty}^{\infty}(-2j-\alpha D)\cos[j\lambda'+(1-j)\lambda-\varpi]b_{1/2}^{(j)} \\
&+ \frac{e'}{2}\sum_{j=-\infty}^{\infty}(-1+2j+\alpha D)\cos[j\lambda'+(1-j)\lambda-\varpi']b_{1/2}^{(j-1)} \\
&- \frac{1}{2}e\alpha\cos(2\lambda-\lambda'-\varpi) + \frac{3}{2}e\alpha\cos(\lambda'-\varpi) - 2e'\alpha\cos(\lambda-2\lambda'+\varpi'),
\end{aligned} \tag{496}$$

$$\begin{aligned}
\mathcal{S}_2 &= \frac{1}{8}(e^2+e'^2)(2\alpha D+\alpha^2D^2)b_{1/2}^{(0)} \\
&+ \frac{ee'}{4}(2-2\alpha D-\alpha^2D^2)b_{1/2}^{(1)}\cos(\varpi-\varpi') \\
&+ ss'\alpha b_{3/2}^{(1)}\cos(\Omega-\Omega'), \\
&- \frac{1}{2}\alpha(s^2+s'^2)b_{3/2}^{(1)}.
\end{aligned} \tag{497}$$

Approximations for the disturbing function of the second body

$$\mathcal{S}' := \mathcal{S}'_0 + \mathcal{S}'_1 + \mathcal{S}'_2 = \alpha \mathcal{S}_D + \alpha^{-1} \mathcal{S}_I, \tag{498}$$

where we expand by order

$$\mathcal{S}'_0 = \frac{1}{2}\alpha \sum_{j=-\infty}^{\infty} b^{(j)}_{1/2} \cos[j(\lambda - \lambda')] - \alpha^{-1} \cos(\lambda - \lambda'), \tag{499}$$

$$\begin{aligned} \mathcal{S}'_1 &= \frac{e}{2}\alpha \sum_{j=-\infty}^{\infty} (-2j - \alpha D) \cos[j\lambda' + (1-j)\lambda - \varpi] b^{(j)}_{1/2} \\ &+ \frac{e'}{2}\alpha \sum_{j=-\infty}^{\infty} (-1 + 2j + \alpha D) \cos[j\lambda' + (1-j)\lambda - \varpi'] b^{(j-1)}_{1/2} \\ &- 2e\alpha^{-1} \cos(\lambda' - 2\lambda + \varpi), -\frac{1}{2} e' \alpha^{-1} \cos(2\lambda' - \lambda - \varpi') + \frac{3}{2} e' \alpha^{-1} \cos(\lambda - \varpi') \end{aligned} \tag{500}$$

$$\begin{aligned} \mathcal{S}'_2 &= \frac{1}{8}\alpha(e^2 + e'^2)(2\alpha D + \alpha^2 D^2) b^{(0)}_{1/2} \\ &+ \frac{ee'}{4}\alpha(2 - 2\alpha D - \alpha^2 D^2) b^{(1)}_{1/2} \cos(\varpi - \varpi') \\ &+ ss'\alpha^2 b^{(1)}_{3/2} \cos(\Omega - \Omega'), \\ &- \frac{1}{2}\alpha^2 (s^2 + s'^2) b^{(1)}_{3/2}. \end{aligned} \tag{501}$$

1.5 Change of Variables

We do substitutions

$$h := e \sin \varpi, \tag{502}$$

$$k := e \cos \varpi, \tag{503}$$

$$p := \sin I \sin \Omega, \tag{504}$$

$$q := \sin I \cos \Omega. \tag{505}$$

This allows us to make arguments of the trigonometric functions in $\mathcal{S}$ and $\mathcal{S}'$ dependent only on λ and λ'. For $\mathcal{S}_1$, we obtain

$$\cos[j\lambda' + (1-j)\lambda - \varpi] = \cos\varpi \cos[j\lambda' + (1-j)\lambda] + \sin\varpi \sin[j\lambda' + (1-j)\lambda] \tag{506}$$

$$= k \cos[j\lambda' + (1-j)\lambda] + h \sin[j\lambda' + (1-j)\lambda], \tag{507}$$

$$e' \cos(\lambda - 2\lambda' + \varpi') = e' \cos\varpi \cos(\lambda - 2\lambda') - e' \sin\varpi \sin(\lambda - 2\lambda') \tag{508}$$

$$= k' \cos(\lambda - 2\lambda') - h' \sin(\lambda - 2\lambda'), \tag{509}$$

$$e \cos(2\lambda - \lambda' - \varpi) = e \cos(\varpi) \cos(2\lambda - \lambda') + e \sin(\varpi) \sin(2\lambda - \lambda') \tag{510}$$

$$= k \cos(2\lambda - \lambda') + h \sin(2\lambda - \lambda'), \tag{511}$$

$$e \cos(\lambda' - \varpi) = e \cos\varpi \cos\lambda' + e \sin\varpi \sin\lambda' \tag{512}$$

$$= k \cos\lambda' + h \sin\lambda'. \tag{513}$$

and can write

$$\mathcal{S}_1 = \frac{1}{2}\sum_{j=-\infty}^{\infty}(-2j-\alpha D)b_{1/2}^{(j)}\left\{k\cos[j\lambda'+(1-j)\lambda]+h\sin[j\lambda'+(1-j)\lambda]\right\} \quad (514)$$
$$+ \frac{1}{2}\sum_{j=-\infty}^{\infty}(-1+2j+\alpha D)b_{1/2}^{(j)}\left\{k'\cos[j\lambda'+(1-j)\lambda]+h'\sin[j\lambda'+(1-j)\lambda]\right\}$$
$$+ \frac{\alpha}{2}\left[-4k'\cos(\lambda-2\lambda')+4h'\sin(\lambda-2\lambda')-k\cos(2\lambda-\lambda')-h\sin(2\lambda-\lambda')+3k\cos\lambda'+3h\sin\lambda'\right],$$

$$\mathcal{S}_1' = \frac{1}{2}\alpha\sum_{j=-\infty}^{\infty}(-2j-\alpha D)b_{1/2}^{(j)}\left\{k\cos[j\lambda'+(1-j)\lambda]+h\sin[j\lambda'+(1-j)\lambda]\right\} \quad (515)$$
$$+ \frac{1}{2}\alpha\sum_{j=-\infty}^{\infty}(-1+2j+\alpha D)b_{1/2}^{(j)}\left\{k'\cos[j\lambda'+(1-j)\lambda]+h'\sin[j\lambda'+(1-j)\lambda]\right\}$$
$$+ \frac{\alpha^{-1}}{2}\left[-4k\cos(\lambda'-2\lambda)+4h\sin(\lambda'-2\lambda)-k'\cos(2\lambda'-\lambda)-h'\sin(2\lambda'-\lambda)+3k'\cos\lambda+3h'\sin\lambda\right].$$

To expand $\mathcal{S}_2$ and $\mathcal{S}_2'$, we note that

$$h^2+k^2+h'^2+k'^2 = e^2\sin^2\varpi+e^2\cos^2\varpi+e'^2\sin^2\varpi'+e'^2\cos^2\varpi' \quad (516)$$
$$= e^2+e'^2, \quad (517)$$
$$p^2+q^2+p'^2+q'^2 = \sin^2 I\sin^2\Omega+\sin^2 I\cos^2\Omega+\sin^2 I'\sin^2\Omega'+\sin^2 I'\cos^2\Omega' \quad (518)$$
$$= \sin^2 I+\sin^2 I' \quad (519)$$
$$= 4(s^2+s'^2), \quad (520)$$
$$pp'+qq' = \sin I\sin I'\sin\Omega\sin\Omega'+\sin I\sin I'\cos\Omega\cos\Omega' \quad (521)$$
$$= \sin I\sin I'\cos(\Omega-\Omega') \quad (522)$$
$$= 4ss'\cos(\Omega-\Omega'), \quad (523)$$
$$hh'+kk' = ee'(\sin\varpi\sin\varpi'+\cos\varpi\cos\varpi') \quad (524)$$
$$= ee'\cos(\varpi-\varpi') \quad (525)$$

to obtain

$$\mathcal{S}_2 = \frac{1}{8}(h^2+k^2+h'^2+k'^2)(2\alpha D+\alpha^2D^2)b_{1/2}^{(0)}+\frac{1}{4}(hh'+kk')(2-2\alpha D-\alpha^2D^2)b_{1/2}^{(1)} \quad (526)$$
$$+ \frac{1}{4}\alpha(pp'+qq')b_{3/2}^{(1)}-\frac{1}{8}\alpha(p^2+q^2+p'^2+q'^2)b_{3/2}^{(1)},$$
$$\mathcal{S}_2' = \frac{1}{8}\alpha(h^2+k^2+h'^2+k'^2)(2\alpha D+\alpha^2D^2)b_{1/2}^{(0)}+\frac{1}{4}\alpha(hh'+kk')(2-2\alpha D-\alpha^2D^2)b_{1/2}^{(1)} \quad (527)$$
$$+ \frac{1}{4}\alpha^2(pp'+qq')b_{3/2}^{(1)}-\frac{1}{8}\alpha^2(p^2+q^2+p'^2+q'^2)b_{3/2}^{(1)}.$$

2 Lagrange Planetary Equations

3 Secular Perturbations

3.1 Two-Planet Solar System

3.1.1 Derivation of Secular Perturbations

First consider two-planet solar system and denote the two planets with subscripts 1 and 2. Then, the disturbing function can be expanded

$$\begin{aligned} \mathcal{S}_D &= \frac{1}{2}\sum_{j=-\infty}^{\infty}\cos[j(\lambda_1-\lambda_2)]b_{1/2}^{(j)} \\ &+ \frac{e_1}{2}(-2j-\alpha D)\cos[j\lambda_2+(1-j)\lambda_1-\varpi_1] \\ &+ \frac{e_2}{2}(-1+2j+\alpha D)\cos[j\lambda_2+(1-j)\lambda_1-\varpi_2] \\ &+ \frac{1}{8}(e_1^2+e_2^2)(2\alpha D+\alpha^2D^2)b_{1/2}^{(0)} \\ &+ \frac{1}{4}e_1e_2(2-2\alpha D-\alpha^2D^2)b_{1/2}^{(1)}\cos(\varpi_1-\varpi_2) \\ &+ s_1s_2\alpha b_{3/2}^{(1)}\cos(\Omega_1-\Omega_2) \\ &- \frac{1}{2}(s_1^2+s_2^2)b_{3/2}^{(1)}\alpha. \end{aligned} \tag{528}$$

We consider only **secular perturbations**, that is perturbations not dependent on the mean longitudes λ_1 and λ_2. Periodic perturbations do not contribute to the long-term averaged behaviour. To further simplify the expression, we also use the results (TODO)

$$(2\alpha D+\alpha^2D^2)b_{1/2}^{(0)} = \alpha b_{3/2}^{(1)} \tag{529}$$

$$(2-2\alpha D-\alpha^2D^2)b_{1/2}^{(1)} = -\alpha b_{3/2}^{(2)} \tag{530}$$

and obtain the purely secular part of the disturbing function

$$\begin{aligned} \mathcal{S}_D^{(sec)} &= \frac{1}{2}b_{1/2}^{(0)}+\frac{1}{8}(e_1^2+e_2^2)\alpha b_{3/2}^{(1)}-\frac{1}{4}e_1e_2\alpha b_{3/2}^{(2)}\cos(\varpi_1-\varpi_2) \\ &+ \frac{1}{4}I_1I_2\alpha b_{3/2}^{(1)}\cos(\Omega_1-\Omega_2)-\frac{1}{8}(I_1^2+I_2^2)\alpha b_{3/2}^{(1)}. \end{aligned} \tag{531}$$

In addition, $\mathcal{S}_I$ and $\mathcal{S}_E$ contain only periodic terms. To obtain the original disturbing functions, we use (17) and (18), to compute

$$\mathcal{R}_1 = \frac{\overline{\mu}_2}{a_2}\mathcal{S}_D^{(sec)} = \frac{Gm_2}{a_2}\mathcal{S}_D^{(sec)} \tag{532}$$

$$\mathcal{R}_2 = \frac{\overline{\mu}_1}{a_1}\alpha\mathcal{S}_D^{(sec)} = \frac{Gm_1}{a_2}\mathcal{S}_D^{(sec)} \tag{533}$$

Kepler's third law

$$\mu_k = n_k^2a_k^3 = G(M+m_k), \tag{534}$$

allows us to eliminate G and obtain

$$\frac{Gm_2}{a_2} = \frac{m_2}{M+m_1}\frac{n_1^2a_1^3}{a_2} = \frac{m_2}{M+m_1}n_1^2a_1^2\alpha \tag{535}$$

$$\frac{Gm_1}{a_2} = \frac{m_1}{M+m_2}\frac{n_2^2a_2^3}{a_2} = \frac{m_1}{M+m_2}n_2^2a_2^2\alpha. \tag{536}$$

Thus, we can express the disturbing functions for the two planets as

$$\mathcal{R}_1 = \frac{m_2}{M+m_1}n_1^2a_1^2\alpha S_D^{(sec)} \tag{537}$$

$$= \frac{m_2}{M+m_1}n_1^2a_1^2\left[\frac{1}{2}\alpha b_{1/2}^{(0)} + \frac{1}{8}(e_1^2+e_2^2)\alpha^2 b_{3/2}^{(1)} - \frac{1}{4}e_1e_2\alpha^2 b_{3/2}^{(2)}\cos(\varpi_1-\varpi_2)\right. \tag{538}$$

$$+ \left.\frac{1}{4}I_1I_2\alpha^2 b_{3/2}^{(1)}\cos(\Omega_1-\Omega_2) - \frac{1}{8}\alpha^2(I_1^2+I_2^2)b_{3/2}^{(1)}\right]$$

$$\mathcal{R}_2 = \frac{m_1}{M+m_2}n_2^2a_2^2S_D^{(sec)} \tag{539}$$

$$= \frac{m_1}{M+m_2}n_2^2a_2^2\left[\frac{1}{2}b_{1/2}^{(0)} + \frac{1}{8}(e_1^2+e_2^2)\alpha b_{3/2}^{(1)} - \frac{1}{4}e_1e_2\alpha b_{3/2}^{(2)}\cos(\varpi_1-\varpi_2)\right. \tag{540}$$

$$+ \left.\frac{1}{4}I_1I_2\alpha b_{3/2}^{(1)}\cos(\Omega_1-\Omega_2) - \frac{1}{8}\alpha(I_1^2+I_2^2)b_{3/2}^{(1)}\right]$$

This expression can be further simplified by discarding terms that do not contribute to the time evolution of orbital elements as described the Lagrange Planetary Equations. In particular, we note that

$$\frac{\partial e_2^2}{\partial e_1} = \frac{\partial e_1^2}{\partial e_2} = \frac{\partial I_2^2}{\partial I_1} = \frac{\partial I_1^2}{\partial I_2} = 0 \tag{541}$$

The derivative w.r.t. a_k is only relevant for the time evolution of mean anomaly, which is not considered here. Thus, also the first term in the square brackets can be discarded. Thus, we obtain

$$\mathcal{R}_1 = \frac{m_2}{M+m_1}n_1^2a_1^2\left[\frac{1}{8}e_1^2\alpha^2 b_{3/2}^{(1)} - \frac{1}{4}e_1e_2\alpha^2 b_{3/2}^{(2)}\cos(\varpi_1-\varpi_2)\right. \tag{542}$$

$$+ \left.\frac{1}{4}I_1I_2\alpha^2 b_{3/2}^{(1)}\cos(\Omega_1-\Omega_2) - \frac{1}{8}\alpha^2 I_1^2 b_{3/2}^{(1)}\right]$$

$$\mathcal{R}_2 = \frac{m_1}{M+m_2}n_2^2a_2^2\left[\frac{1}{8}e_2^2\alpha b_{3/2}^{(1)} - \frac{1}{4}e_1e_2\alpha b_{3/2}^{(2)}\cos(\varpi_1-\varpi_2)\right. \tag{543}$$

$$+ \left.\frac{1}{4}I_1I_2\alpha b_{3/2}^{(1)}\cos(\Omega_1-\Omega_2) - \frac{1}{8}\alpha I_2^2 b_{3/2}^{(1)}\right]$$

Let us denote $\alpha_{12} := \alpha = a_1/a_2$

$$A_{jj} := \frac{1}{4}n_j\frac{m_k}{M+m_j}\alpha_{12}\overline{\alpha}_{12}b_{3/2}^{(1)}(\alpha_{12}), \tag{544}$$

$$A_{jk} := -\frac{1}{4}n_j\frac{m_k}{M+m_j}\alpha_{12}\overline{\alpha}_{12}b_{3/2}^{(2)}(\alpha_{12}), \tag{545}$$

$$B_{jj} := -\frac{1}{4}n_j\frac{m_k}{M+m_j}\alpha_{12}\overline{\alpha}_{12}b_{3/2}^{(1)}(\alpha_{12}), \tag{546}$$

$$B_{jk} := \frac{1}{4}n_j\frac{m_k}{M+m_j}\alpha_{12}\overline{\alpha}_{12}b_{3/2}^{(1)}(\alpha_{12}), \tag{547}$$

where $k \neq j$ and

$$\overline{\alpha}_{12} = \begin{cases} \alpha_{12}, & j = 1 \\ 1, & j = 2 \end{cases} . \tag{548}$$

Then, the disturbing functions obtain more convenient expression

$$\mathcal{R}_j = n_j a_j^2 \left[\frac{1}{2} A_{jj} e_j^2 + A_{jk} e_1 e_2 \cos(\varpi_1 - \varpi_2) + \frac{1}{2} B_{jj} I_j^2 + B_{jk} I_1 I_2 \cos(\Omega_1 - \Omega_2) \right]. \tag{549}$$

Let's again perform change of variables

$$h_j = e_j \sin \varpi_j, \tag{550}$$
$$k_j = e_j \cos \varpi_j, \tag{551}$$
$$p_j = I_j \sin \Omega_j, \tag{552}$$
$$q_j = I_j \cos \Omega_j. \tag{553}$$

Then, we obtain

$$h_j^2 + k_j^2 = e_j^2(\sin^2 \varpi_j + \cos^2 \varpi_j) = e_j^2 \tag{554}$$
$$p_j^2 + q_j^2 = I_j^2(\sin^2 \Omega_j + \cos^2 \Omega_j) = I_j^2 \tag{555}$$
$$h_1 h_2 + k_1 k_2 = e_1 e_2(\sin \varpi_1 \sin \varpi_2 + \cos \varpi_1 \cos \varpi_2) = e_1 e_2 \cos(\varpi_1 - \varpi_2) \tag{556}$$
$$p_1 p_2 + q_1 q_2 = I_1 I_2(\sin \Omega_1 \sin \Omega_2 + \cos \Omega_1 \cos \Omega_2) = I_1 I_2 \cos(\Omega_1 - \Omega_2) \tag{557}$$

and

$$\mathcal{R}_j = n_j a_j^2 \left[\frac{1}{2} A_{jj}(h_j^2 + k_j^2) + A_{jk}(h_1 h_2 + k_1 k_2) + \frac{1}{2} B_{jj}(p_j^2 + q_j^2) + B_{jk}(p_1 p_2 + q_1 q_2) \right]. \tag{558}$$

3.1.2 Approximation of the Planetary Equations

We employ first-order approximation of Lagrange's planetary equations

$$\frac{de}{dt} = -\frac{\sqrt{1-e^2}}{na^2 e}\left(1 - \sqrt{1-e^2}\right)\frac{d\mathcal{R}}{d\epsilon} - \frac{\sqrt{1-e^2}}{na^2 e}\frac{\partial \mathcal{R}}{\partial \varpi} \approx \frac{-1}{na^2 e}\frac{\partial \mathcal{R}}{\partial \varpi}, \tag{559}$$
$$\frac{d\Omega}{dt} = \frac{1}{na^2\sqrt{1-e^2}\sin I}\frac{\partial \mathcal{R}}{\partial I} \approx \frac{1}{na^2 I}\frac{\partial \mathcal{R}}{\partial I}, \tag{560}$$
$$\frac{d\varpi}{dt} = \frac{\sqrt{1-e^2}}{na^2 e}\frac{\partial \mathcal{R}}{\partial e} + \frac{\tan(I/2)}{na^2\sqrt{1-e^2}}\frac{\partial \mathcal{R}}{\partial I} \approx \frac{1}{na^2 e}\frac{\partial \mathcal{R}}{\partial e}, \tag{561}$$
$$\frac{\partial I}{\partial t} = -\frac{\tan(I/2)}{na^2\sqrt{1-e^2}}\left(\frac{\partial \mathcal{R}}{\partial \epsilon} + \frac{\partial \mathcal{R}}{\partial \varpi}\right) - \frac{1}{na^2\sqrt{1-e^2}\sin I}\frac{\partial \mathcal{R}}{\partial \Omega} \approx \frac{-1}{na^2 I}\frac{\partial \mathcal{R}}{\partial \Omega}. \tag{562}$$

In order to compute the time derivatives, we need to compute the time partial derivatives of the disturbing function

$$\frac{\partial \mathcal{R}_j}{\partial \varpi_j} = n_j a_j^2 \left[A_{jk} e_1 e_2 \frac{\partial}{\partial \varpi_j} \cos(\varpi_1 - \varpi_2) \right] \tag{563}$$
$$= n_j a_j^2 A_{jk} e_1 e_2 (-1)^j \sin(\varpi_1 - \varpi_2) \tag{564}$$

$$\frac{\partial \mathcal{R}_j}{\partial I_j} = n_j a_j^2 \left[B_{jj} I_j + B_{jk} I_k \cos(\Omega_1 - \Omega_2) \right] \tag{565}$$

$$= n_j a_j^2 B_{jj} I_j + n_j a_j^2 B_{jk} I_k \cos(\Omega_1 - \Omega_2), \tag{566}$$

$$\frac{\partial \mathcal{R}_j}{\partial e_j} = n_j a_j^2 \left[A_{jj} e_j + A_{jk} e_k \cos(\varpi_1 - \varpi_2) \right] \tag{567}$$

$$= n_j a_j^2 A_{jj} e_j + n_j a_j^2 A_{jk} e_k \cos(\varpi_1 - \varpi_2), \tag{568}$$

$$\frac{\partial \mathcal{R}_j}{\partial \Omega_j} = n_j a_j^2 B_{jk} I_1 I_2 \frac{\partial}{\partial \Omega_j} \cos(\Omega_1 - \Omega_2) \tag{569}$$

$$= n_j a_j^2 B_{jk} I_1 I_2 (-1)^j \sin(\Omega_1 - \Omega_2). \tag{570}$$

Substituting back to the planetary equations, we obtain

$$\frac{de_j}{dt} = e_k A_{jk} (-1)^k \sin(\varpi_1 - \varpi_2), \tag{571}$$

$$\frac{d\Omega_j}{dt} = B_{jj} + B_{jk} \frac{I_k}{I_j} \cos(\Omega_1 - \Omega_2), \tag{572}$$

$$\frac{d\varpi_j}{dt} = A_{jj} + A_{jk} \frac{e_k}{e_j} \cos(\varpi_1 - \varpi_2), \tag{573}$$

$$\frac{dI_j}{dt} = I_k B_{jk} (-1)^k \sin(\Omega_1 - \Omega_2). \tag{574}$$

Since eccentricity and inclination are small, their presence in the denominators can be problematic and suggests that a change of variables is appropriate. First note that

$$\frac{\partial \mathcal{R}_j}{\partial e_j} = \frac{\partial \mathcal{R}_j}{\partial h_j}\frac{\partial h_j}{\partial e_j} + \frac{\partial \mathcal{R}_j}{\partial k_j}\frac{\partial k_j}{\partial e_j} = \frac{h_j}{e_j}\frac{\partial \mathcal{R}_j}{\partial h_j} + \frac{k_j}{e_j}\frac{\partial \mathcal{R}_j}{\partial k_j}, \tag{575}$$

$$\frac{\partial \mathcal{R}_j}{\partial \varpi_j} = \frac{\partial \mathcal{R}_j}{\partial h_j}\frac{\partial h_j}{\partial \varpi_j} + \frac{\partial \mathcal{R}_j}{\partial k_j}\frac{\partial k_j}{\partial \varpi_j} = k_j \frac{\partial \mathcal{R}_j}{\partial h_j} - h_j \frac{\partial \mathcal{R}_j}{\partial k_j}, \tag{576}$$

$$\frac{\partial \mathcal{R}_j}{\partial I_j} = \frac{\partial \mathcal{R}_j}{\partial p_j}\frac{\partial p_j}{\partial I_j} + \frac{\partial \mathcal{R}_j}{\partial q_j}\frac{\partial q_j}{\partial I_j} = \frac{p_j}{I_j}\frac{\partial \mathcal{R}_j}{\partial p_j} + \frac{q_j}{I_j}\frac{\partial \mathcal{R}_j}{\partial q_j}, \tag{577}$$

$$\frac{\partial \mathcal{R}_j}{\partial \Omega_j} = \frac{\partial \mathcal{R}_j}{\partial p_j}\frac{\partial p_j}{\partial \Omega_j} + \frac{\partial \mathcal{R}_j}{\partial q_j}\frac{\partial q_j}{\partial \Omega_j} = q_j \frac{\partial \mathcal{R}_j}{\partial p_j} - p_j \frac{\partial \mathcal{R}_j}{\partial q_j}. \tag{578}$$

Now we can derive the approximate planetary equations for the new variables as follows

$$\frac{dh_j}{dt} = \frac{\partial h_j}{\partial e_j}\frac{de_j}{dt} + \frac{\partial h_j}{\partial \varpi_j}\frac{d\varpi_j}{dt} \tag{579}$$

$$= \frac{h_j}{e_j}\left(\frac{-1}{n_j a_j^2 e_j}\frac{\partial \mathcal{R}_j}{\partial \varpi_j}\right) + k_j\left(\frac{1}{n_j a_j^2 e_j}\frac{\partial \mathcal{R}_j}{\partial e_j}\right) \tag{580}$$

$$= \frac{1}{n_j a_j^2 e_j}\left(-\frac{h_j}{e_j}\frac{\partial \mathcal{R}_j}{\partial \varpi_j} + k_j\frac{\partial \mathcal{R}_j}{\partial e_j}\right) \tag{581}$$

$$= \frac{1}{n_j a_j^2 e_j}\left[-\frac{h_j}{e_j}\left(k_j\frac{\partial \mathcal{R}_j}{\partial h_j} - h_j\frac{\partial \mathcal{R}_j}{\partial k_j}\right) + k_j\left(\frac{h_j}{e_j}\frac{\partial \mathcal{R}_j}{\partial h_j} + \frac{k_j}{e_j}\frac{\partial \mathcal{R}_j}{\partial k_j}\right)\right] \tag{582}$$

$$= \frac{1}{n_j a_j^2 e_j}\left[\frac{\partial \mathcal{R}_j}{\partial h_j}\left(-\frac{h_j k_j}{e_j} + \frac{h_j k_j}{e_j}\right) + \frac{\partial \mathcal{R}_j}{\partial k_j}\left(\frac{h_j^2}{e_j} + \frac{k_j^2}{e_j}\right)\right] \tag{583}$$

$$= \frac{1}{n_j a_j^2}\frac{\partial \mathcal{R}_j}{\partial k_j}, \tag{584}$$

$$\frac{dk_j}{dt} = \frac{\partial k_j}{\partial e_j}\frac{de_j}{dt} + \frac{\partial k_j}{\partial \varpi_j}\frac{d\varpi_j}{dt} \tag{585}$$

$$= \frac{k_j}{e_j}\left(\frac{-1}{n_j a_j^2 e_j}\frac{\partial \mathcal{R}_j}{\partial \varpi_j}\right) - h_j\left(\frac{1}{n_j a_j^2 e_j}\frac{\partial \mathcal{R}_j}{\partial e_j}\right) \tag{586}$$

$$= \frac{1}{n_j a_j^2 e_j}\left[-\frac{k_j}{e_j}\left(k_j\frac{\partial \mathcal{R}_j}{\partial h_j} - h_j\frac{\partial \mathcal{R}_j}{\partial k_j}\right) - h_j\left(\frac{h_j}{e_j}\frac{\partial \mathcal{R}_j}{\partial h_j} + \frac{k_j}{e_j}\frac{\partial \mathcal{R}_j}{\partial k_j}\right)\right] \tag{587}$$

$$= \frac{1}{n_j a_j^2 e_j}\left[\frac{\partial \mathcal{R}_j}{\partial h_j}\left(-\frac{k_j^2}{e_j} - \frac{h_j^2}{e_j}\right) + \frac{\partial \mathcal{R}_j}{\partial k_j}\left(\frac{k_j h_j}{e_j} - \frac{h_j k_j}{e_j}\right)\right] \tag{588}$$

$$= -\frac{1}{n_j a_j^2}\frac{\partial \mathcal{R}_j}{\partial h_j}, \tag{589}$$

$$\frac{dp_j}{dt} = \frac{\partial p_j}{\partial I_j}\frac{dI_j}{dt} + \frac{\partial p_j}{\partial \Omega_j}\frac{d\Omega_j}{dt} \tag{590}$$

$$= \frac{p_j}{I_j}\left(\frac{-1}{n_j a_j^2 I_j}\frac{\partial \mathcal{R}_j}{\partial \Omega_j}\right) + q_j\left(\frac{1}{n_j a_j^2 I_j}\frac{\partial \mathcal{R}_j}{\partial I_j}\right) \tag{591}$$

$$= \frac{1}{n_j a_j^2 I_j}\left[-\frac{p_j}{I_j}\left(q_j\frac{\partial \mathcal{R}_j}{\partial p_j} - p_j\frac{\partial \mathcal{R}_j}{\partial q_j}\right) + q_j\left(\frac{p_j}{I_j}\frac{\partial \mathcal{R}_j}{\partial p_j} + \frac{q_j}{I_j}\frac{\partial \mathcal{R}_j}{\partial q_j}\right)\right] \tag{592}$$

$$= \frac{1}{n_j a_j^2 I_j}\left[\frac{\partial \mathcal{R}_j}{\partial p_j}\left(-\frac{p_j q_j}{I_j} + \frac{q_j p_j}{I_j}\right) + \frac{\partial \mathcal{R}_j}{\partial q_j}\left(\frac{p_j^2}{I_j} + \frac{q_j^2}{I_j}\right)\right] \tag{593}$$

$$= \frac{1}{n_j a_j^2}\frac{\partial \mathcal{R}_j}{\partial q_j}, \tag{594}$$

$$\frac{dq_j}{dt} = \frac{\partial q_j}{\partial I_j}\frac{dI_j}{dt} + \frac{\partial q_j}{\partial \Omega_j}\frac{d\Omega_j}{dt} \tag{595}$$

$$= \frac{q_j}{I_j}\frac{dI_j}{dt} - p_j\frac{d\Omega_j}{dt} \tag{596}$$

$$= \frac{q_j}{I_j}\left[\frac{-1}{n_j a_j^2 I_j}\left(q_j\frac{\partial \mathcal{R}_j}{\partial p_j} - p_j\frac{\partial \mathcal{R}_j}{\partial q_j}\right)\right] - p_j\left[\frac{1}{n_j a_j^2 I_j}\left(\frac{p_j}{I_j}\frac{\partial \mathcal{R}_j}{\partial p_j} + \frac{q_j}{I_j}\frac{\partial \mathcal{R}_j}{\partial q_j}\right)\right] \tag{597}$$

$$= \frac{1}{n_j a_j^2 I_j}\left[\frac{\partial \mathcal{R}_j}{\partial p_j}\left(-\frac{q_j^2}{I_j} - \frac{p_j^2}{I_j}\right) + \frac{\partial \mathcal{R}_j}{\partial q_j}\left(\frac{p_j q_j}{I_j} - \frac{p_j q_j}{I_j}\right)\right] \tag{598}$$

$$= \frac{-1}{n_j a_j^2}\frac{\partial \mathcal{R}_j}{\partial p_j}. \tag{599}$$

That is, collecting the result yields

$$\frac{dh_j}{dt} = \frac{1}{n_j a_j^2}\frac{\partial \mathcal{R}_j}{\partial k_j} = A_{jj}k_j + A_{jk}k_k, \tag{600}$$

$$\frac{dk_j}{dt} = -\frac{1}{n_j a_j^2}\frac{\partial \mathcal{R}_j}{\partial h_j} = -A_{jj}h_j - A_{jk}h_k, \tag{601}$$

$$\frac{dp_j}{dt} = \frac{1}{n_j a_j^2}\frac{\partial \mathcal{R}_j}{\partial q_j} = B_{jj}q_j + B_{jk}q_k, \tag{602}$$

$$\frac{dq_j}{dt} = -\frac{1}{n_j a_j^2}\frac{\partial \mathcal{R}_j}{\partial p_j} = -B_{jj}p_j - B_{jk}p_k. \tag{603}$$

3.1.3 Laplace-Lagrange Secular Solution

The two-planet system is now described by the systems of differential equations

$$\dot{\boldsymbol{h}} = A\boldsymbol{k}, \tag{604}$$

$$\dot{\boldsymbol{k}} = -A\boldsymbol{h}, \tag{605}$$

$$\dot{\boldsymbol{p}} = B\boldsymbol{q}, \tag{606}$$

$$\dot{\boldsymbol{q}} = -B\boldsymbol{p}. \tag{607}$$

Let $\boldsymbol{u}_1, \boldsymbol{u}_2$ be the eigenvectors corresponding to the eigenvalues g_1, g_2 for the matrix A. Let's try

$$\boldsymbol{h}(t) = \boldsymbol{u}_1 \sin(g_1 t + \beta_1) + \boldsymbol{u}_2 \sin(g_2 t + \beta_2), \tag{608}$$

$$\boldsymbol{k}(t) = \boldsymbol{u}_1 \cos(g_1 t + \beta_1) + \boldsymbol{u}_2 \cos(g_2 t + \beta_2). \tag{609}$$

Then, we obtain

$$\begin{aligned}
\dot{\boldsymbol{h}}(t) &= g_1\boldsymbol{u}_1\cos(g_1t+\beta_1)+g_2\boldsymbol{u}_2\cos(g_2t+\beta_2) && (610)\\
&= A\boldsymbol{u}_1\cos(g_1t+\beta_1)+A\boldsymbol{u}_2\cos(g_2t+\beta_2) && (611)\\
&= A\boldsymbol{k}(t), && (612)\\
\dot{\boldsymbol{k}}(t) &= -g_1\boldsymbol{u}_1\sin(g_1t+\beta_1)-g_2\boldsymbol{u}_2\sin(g_2t+\beta_2) && (613)\\
&= -A\boldsymbol{u}_1\sin(g_1t+\beta_1)-A\boldsymbol{u}_2\sin(g_2t+\beta_2) && (614)\\
&= -A\boldsymbol{h}(t), && (615)
\end{aligned}$$

which shows that the solution is correct. Similarly, let $\boldsymbol{w}_1, \boldsymbol{w}_2$ be the eigenvectors corresponding to the eigenvalues f_1, f_2 for the matrix $\boldsymbol{B}$. This, leads to the solutions

$$\begin{aligned}
\boldsymbol{p}(t) &= \boldsymbol{w}_1\sin(f_1t+\gamma_1)+\boldsymbol{w}_2\sin(f_2t+\gamma_2), && (616)\\
\boldsymbol{q}(t) &= \boldsymbol{w}_1\cos(f_1t+\gamma_1)+\boldsymbol{w}_2\cos(f_2t+\gamma_2). && (617)
\end{aligned}$$

Note that for any eigenvector, the eigenvector multiplied by a non-zero scalar is also an eigenvector. The scaling factors for $\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{w}_1, \boldsymbol{w}_2$ and the coefficients $\beta_1, \beta_2, \gamma_1, \gamma_2$ follow from the $4+4$ initial conditions for h_1, k_1, p_1, q_1 and h_2, k_2, p_2, q_2. This solution is an instance of **classical Laplace-Lagrange secular solution**.

3.1.4 Example: Jupiter and Saturn

Suppose the two planets are Jupiter and Saturn with mass ratios and initial conditions

$$\begin{array}{llll}
m_1/M = & 9.54786\cdot10^{-4}, & m_2/M = & 2.85837\cdot10^{-4}\\
a_1 = & 5.202545\ \text{AU}, & a_2 = & 9.554841\ \text{AU}\\
n_1 = & 30.3374°/y, & n_2 = & 12.1890°/y\\
e_1 = & 0.0474622, & e_2 = & 0.0575481\\
\varpi_1 = & 13.983865°, & \varpi_2 = & 88.719425°\\
I_1 = & 1.30667°, & I_2 = & 2.48795°\\
\Omega_1 = & 100.0381°, & \Omega_2 = & 113.1334°.
\end{array} \quad (618)$$

This implies $\alpha_{12} = 0.5444931004084735$ and we obtain via numerical integration

$$\begin{aligned}
b^{(1)}_{3/2} &= 3.172964573849452, && (619)\\
b^{(2)}_{3/2} &= 2.0710972389662508. && (620)
\end{aligned}$$

Using the solar mass $M = 1.988416\cdot10^{30}$ kg, yields $m_1 = 1.898511758976\cdot10^{27}$ kg and $m_2 = 5.68362864192\cdot10^{26}$ kg and

$$\begin{aligned}
A_{11} &= \frac{1}{4}n_1\frac{m_2}{M+m_1}\alpha_{12}^2 b^{(1)}_{3/2} = 0.002037381396074496°/y, && (621)\\
A_{22} &= \frac{1}{4}n_2\frac{m_1}{M+m_2}\alpha_{12} b^{(1)}_{3/2} = 0.005025133599532606°/y, && (622)\\
A_{12} &= -\frac{1}{4}n_1\frac{m_2}{M+m_1}\alpha_{12}^2 b^{(2)}_{3/2} = -0.001329865142178957°/y, && (623)\\
A_{21} &= -\frac{1}{4}n_2\frac{m_1}{M+m_2}\alpha_{12} b^{(2)}_{3/2} = -0.003280068239401 1267°/y. && (624)
\end{aligned}$$

$$B_{11} = -\frac{1}{4}n_1\frac{m_2}{M+m_1}\alpha_{12}^2 b_{3_2}^{(1)} = -0.002037381396074496, \quad (625)$$

$$B_{22} = -\frac{1}{4}n_2\frac{m_1}{M+m_2}\alpha_{12} b_{3_2}^{(1)} = -0.005025133599532606, \quad (626)$$

$$B_{12} = \frac{1}{4}n_1\frac{m_2}{M+m_1}\alpha_{12}^2 b_{3_2}^{(1)} = 0.002037381396074496, \quad (627)$$

$$B_{21} = \frac{1}{4}n_2\frac{m_1}{M+m_2}\alpha_{12} b_{3_2}^{(1)} = 0.005025133599532606. \quad (628)$$

Thus, we obtain the matrices

$$A = \begin{bmatrix} 0.00203738 & -0.00132987 \\ -0.00328007 & 0.00502513 \end{bmatrix} {}^\circ/y, \quad (629)$$

$$B = \begin{bmatrix} -0.00203738 & 0.00203738 \\ 0.00502513 & -0.00502513 \end{bmatrix} {}^\circ/y. \quad (630)$$

The eigenvalues and eigenvectors obtain the values

$$g_1 = 0.0009634346412059512, \quad (631)$$

$$g_2 = 0.006099080354401151, \quad (632)$$

$$\overline{\boldsymbol{u}}_1 = \begin{bmatrix} -0.77799125 & -0.62827511 \end{bmatrix}^T \quad (633)$$

$$\overline{\boldsymbol{u}}_2 = \begin{bmatrix} 0.31116203 & -0.95035688 \end{bmatrix}^T, \quad (634)$$

$$f_1 = 0 \quad (635)$$

$$f_2 = -0.007062514995607103, \quad (636)$$

$$\overline{\boldsymbol{w}}_1 = \begin{bmatrix} 0.70710678 & 0.70710678 \end{bmatrix}^T, \quad (637)$$

$$\overline{\boldsymbol{w}}_2 = \begin{bmatrix} -0.37573125 & 0.92672867 \end{bmatrix}^T. \quad (638)$$

Let S_1, S_2, T_1, T_2 be scalars determined by initial conditions such that

$$\boldsymbol{u}_k = S_k\overline{\boldsymbol{u}}_k, \quad (639)$$

$$\boldsymbol{w}_k = T_k\overline{\boldsymbol{w}}_k. \quad (640)$$

Requiring that the solution agrees with the initial condition at $t = 0$ implies

$$h_1(0) = u_{11}\sin\beta_1 + u_{12}\sin\beta_2 = S_1\overline{u}_{11}\sin\beta_1 + S_2\overline{u}_{12}\sin\beta_2 = e_1\sin\varpi_1 = 0.01146918, \quad (641)$$

$$h_2(0) = u_{21}\sin\beta_1 + u_{22}\sin\beta_2 = S_1\overline{u}_{21}\sin\beta_1 + S_2\overline{u}_{22}\sin\beta_2 = e_2\sin\varpi_2 = 0.05753373, \quad (642)$$

$$k_1(0) = u_{11}\cos\beta_1 + u_{12}\cos\beta_2 = S_1\overline{u}_{11}\cos\beta_1 + S_2\overline{u}_{12}\cos\beta_2 = e_1\cos\varpi_1 = 0.04605560, \quad (643)$$

$$k_2(0) = u_{21}\cos\beta_1 + u_{22}\cos\beta_2 = S_1\overline{u}_{21}\cos\beta_1 + S_2\overline{u}_{22}\cos\beta_2 = e_2\cos\varpi_2 = 0.001286107, \quad (644)$$

$$p_1(0) = w_{11}\sin\gamma_1 + w_{12}\sin\gamma_2 = T_1\overline{w}_{11}\sin\gamma_1 + T_2\overline{w}_{12}\sin\gamma_2 = I_1\sin\Omega_1 = 0.02245659, \quad (645)$$

$$p_2(0) = w_{21}\sin\gamma_1 + w_{22}\sin\gamma_2 = T_1\overline{w}_{21}\sin\gamma_1 + T_2\overline{w}_{22}\sin\gamma_2 = I_2\sin\Omega_2 = 0.03993140, \quad (646)$$

$$q_1(0) = w_{11}\cos\gamma_1 + w_{12}\cos\gamma_2 = T_1\overline{w}_{11}\cos\gamma_1 + T_2\overline{w}_{12}\cos\gamma_2 = I_1\cos\Omega_1 = -0.00397510, \quad (647)$$

$$q_2(0) = w_{21}\cos\gamma_1 + w_{22}\cos\gamma_2 = T_1\overline{w}_{21}\cos\gamma_1 + T_2\overline{w}_{22}\cos\gamma_2 = I_2\cos\Omega_2 = -0.01705970. \quad (648)$$

These equations can be expressed as a set of two linear systems of equations

$$\begin{bmatrix} \overline{u}_{11} & \overline{u}_{12} & 0 & 0 \\ \overline{u}_{21} & \overline{u}_{22} & 0 & 0 \\ 0 & 0 & \overline{u}_{11} & \overline{u}_{12} \\ 0 & 0 & \overline{u}_{21} & \overline{u}_{22} \end{bmatrix} \begin{bmatrix} S_1 \sin\beta_1 \\ S_2 \sin\beta_2 \\ S_1 \cos\beta_1 \\ S_2 \cos\beta_2 \end{bmatrix} = \begin{bmatrix} e_1 \sin\varpi_1 \\ e_2 \sin\varpi_2 \\ e_1 \cos\varpi_1 \\ e_2 \cos\varpi_2 \end{bmatrix} \tag{649}$$

$$\begin{bmatrix} \overline{w}_{11} & \overline{w}_{12} & 0 & 0 \\ \overline{w}_{21} & \overline{w}_{22} & 0 & 0 \\ 0 & 0 & \overline{w}_{11} & \overline{w}_{12} \\ 0 & 0 & \overline{w}_{21} & \overline{w}_{22} \end{bmatrix} \begin{bmatrix} T_1 \sin\gamma_1 \\ T_2 \sin\gamma_2 \\ T_1 \cos\gamma_1 \\ T_2 \cos\gamma_2 \end{bmatrix} = \begin{bmatrix} I_1 \sin\Omega_1 \\ I_2 \sin\Omega_2 \\ I_1 \cos\Omega_1 \\ I_2 \cos\Omega_2 \end{bmatrix} \tag{650}$$

which can be solved

$$\begin{bmatrix} S_1 \sin\beta_1 \\ S_2 \sin\beta_2 \\ S_1 \cos\beta_1 \\ S_2 \cos\beta_2 \end{bmatrix} = \begin{bmatrix} -0.77799125 & 0.31116203 & 0 & 0 \\ -0.62827511 & -0.95035688 & 0 & 0 \\ 0 & 0 & -0.77799125 & 0.31116203 \\ 0 & 0 & -0.62827511 & -0.95035688 \end{bmatrix}^{-1} \begin{bmatrix} 0.01146918 \\ 0.05753373 \\ 0.04605560 \\ 0.00128611 \end{bmatrix} \tag{651}$$

$$\begin{bmatrix} T_1 \sin\gamma_1 \\ T_2 \sin\gamma_2 \\ T_1 \cos\gamma_1 \\ T_2 \cos\gamma_2 \end{bmatrix} = \begin{bmatrix} 0.70710678 & -0.37573125 & 0 & 0 \\ 0.70710678 & 0.92672867 & 0 & 0 \\ 0 & 0 & 0.70710678 & -0.37573125 \\ 0 & 0 & 0.70710678 & 0.92672867 \end{bmatrix}^{-1} \begin{bmatrix} 0.02245659 \\ 0.03993140 \\ -0.00397510 \\ -0.01705970 \end{bmatrix} \tag{652}$$

leading to

$$\begin{array}{llll} S_1 \sin\beta_1 = & -0.03080887, & T_1 \sin\gamma_1 = & 0.03888761, \\ S_2 \sin\beta_2 = & -0.04017152, & T_2 \sin\gamma_2 = & 0.01341677, \\ S_1 \cos\beta_1 = & -0.04724689, & T_1 \cos\gamma_1 = & -0.01095976, \\ S_2 \cos\beta_2 = & 0.02988133, & T_2 \cos\gamma_2 = & -0.01004607. \end{array} \tag{653}$$

Thus, finally

$$S_1 = \sqrt{(S_1 \sin\beta_1)^2 + (S_1 \cos\beta_1)^2} = 0.05640439, \tag{654}$$

$$S_2 = \sqrt{(S_2 \sin\beta_2)^2 + (S_2 \cos\beta_2)^2} = 0.05006640, \tag{655}$$

$$T_1 = \sqrt{(T_1 \sin\gamma_1)^2 + (T_1 \cos\gamma_1)^2} = 0.04040251, \tag{656}$$

$$T_2 = \sqrt{(T_2 \sin\gamma_2)^2 + (T_2 \cos\gamma_2)^2} = 0.01676106, \tag{657}$$

$$\beta_1 = \text{atan2}(S_1 \sin\beta_1, S_1 \cos\beta_1) = -146.89233296°, \tag{658}$$

$$\beta_2 = \text{atan2}(S_2 \sin\beta_2, S_1 \cos\beta_2) = -53.35651935°, \tag{659}$$

$$\gamma_1 = \text{atan2}(T_1 \sin\gamma_1, T_1 \cos\gamma_1) = 105.73951730°, \tag{660}$$

$$\gamma_2 = \text{atan2}(T_2 \sin\gamma_2, T_2 \cos\gamma_2) = 126.82475444°. \tag{661}$$

and we can express the eigenvectors

$$\begin{aligned}
\boldsymbol{u}_1 &= S_1\overline{\boldsymbol{u}}_1 = \begin{bmatrix}-0.04388212 & -0.03543747\end{bmatrix}^T, && (662)\\
\boldsymbol{u}_2 &= S_2\overline{\boldsymbol{u}}_2 = \begin{bmatrix}0.01557876 & -0.04758095\end{bmatrix}^T, && (663)\\
\boldsymbol{w}_1 &= T_1\overline{\boldsymbol{w}}_1 = \begin{bmatrix}0.02856889 & 0.02856889\end{bmatrix}^T, && (664)\\
\boldsymbol{w}_2 &= T_2\overline{\boldsymbol{w}}_2 = \begin{bmatrix}-0.00629765 & 0.01553295\end{bmatrix}^T. && (665)
\end{aligned}$$

Now we finally can express the solution

$$\begin{aligned}
h_1(t) &= u_{11}\sin(g_1t+\beta_1) + u_{12}\sin(g_2t+\beta_2) && (666)\\
&- 0.04388212\sin(0.0009634346°/y\cdot t - 146.89233296°) && (667)\\
&+ 0.01557876\sin(0.0060990804°/y\cdot t - 53.35651935°), \\
h_2(t) &= u_{21}\sin(g_1t+\beta_1) + u_{22}\sin(g_2t+\beta_2) && (668)\\
&- 0.03543747\sin(0.0009634346°/y\cdot t - 146.89233296°) && (669)\\
&- 0.04758095\sin(0.0060990804°/y\cdot t - 53.35651935°), \\
k_1(t) &= u_{11}\cos(g_1t+\beta_1) + u_{12}\cos(g_2t+\beta_2) && (670)\\
&- 0.04388212\cos(0.0009634346°/y\cdot t - 146.89233296°) && (671)\\
&+ 0.01557876\cos(0.0060990804°/y\cdot t - 53.35651935°), \\
k_2(t) &= u_{21}\cos(g_1t+\beta_1) + u_{22}\cos(g_2t+\beta_2) && (672)\\
&- 0.03543747\cos(0.0009634346°/y\cdot t - 146.89233296°) && (673)\\
&- 0.04758095\cos(0.0060990804°/y\cdot t - 53.35651935°), \\
p_1(t) &= w_{11}\sin(f_1t+\gamma_1) + w_{12}\sin(f_2t+\gamma_2) && (674)\\
&= 0.02856889\sin(105.73951730°) && (675)\\
&- 0.00629765\sin(0.007062515°/y\cdot t + 126.8247544°), \\
p_2(t) &= w_{21}\sin(f_1t+\gamma_1) + w_{22}\sin(f_2t+\gamma_2) && (676)\\
&= 0.02856889\sin(105.73951730°) && (677)\\
&+ 0.01553295\sin(0.007062515°/y\cdot t + 126.8247544°), \\
q_1(t) &= w_{11}\cos(f_1t+\gamma_1) + w_{12}\cos(f_2t+\gamma_2) && (678)\\
&= 0.02856889\cos(105.73951730°) && (679)\\
&- 0.00629765\cos(0.007062515°/y\cdot t + 126.8247544°), \\
q_2(t) &= w_{21}\cos(f_1t+\gamma_1) + w_{22}\cos(f_2t+\gamma_2) && (680)\\
&= 0.02856889\cos(105.73951730°) && (681)\\
&+ 0.01553295\cos(0.007062515°/y\cdot t + 126.8247544°).
\end{aligned}$$

The Keplerian elements computed from the solution are shown in Figure (1).

3.2 Generalized Secular Perturbations

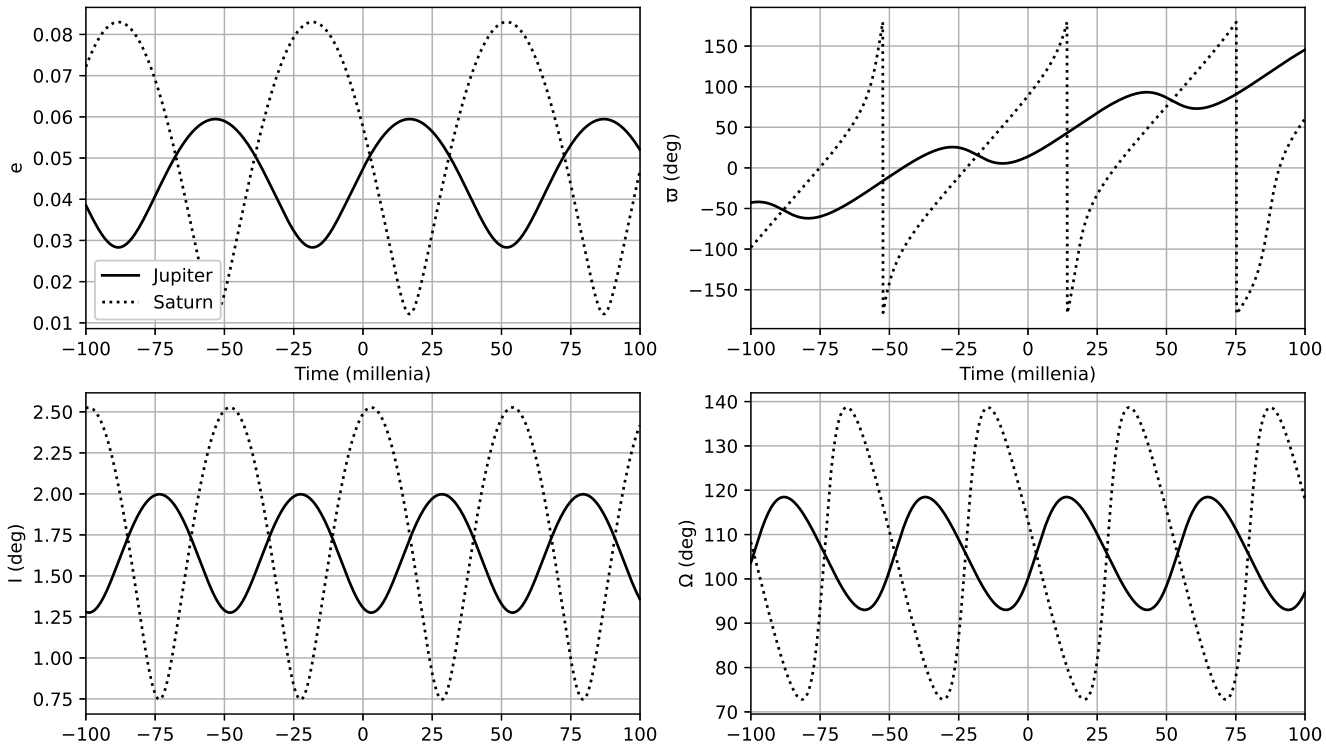


Figure 1: Evolution of Keplerian elements from two-planet secular perturbation theory.

A Series Expansions

In this section, we develop polynomial approximations for E, $\sin E$, $\cos E$, $\sin f$, $\cos f$ in terms of the mean anomaly M and eccentricity e. First, series expansions will be derived as Fourier series expansion w.r.t. M with coefficients that involve Bessel functions and eccentricity. Then, using polynomial approximation for the Bessel functions, the expressions will be expanded to a finite order of e.

Any sufficiently well-behaved function $g(M)$ defined on the interval $[0, 2\pi]$ can be expanded as a **Fourier series**

$$g(M) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_n \cos kM \; dx + \sum_{k=1}^{\infty} b_n \sin kM \; dx, \tag{682}$$

where the Fourier series coefficients are defined

$$a_0 := \frac{1}{2\pi} \int_0^{2\pi} g(M) \, dM, \tag{683}$$

$$a_k := \frac{1}{\pi} \int_0^{2\pi} g(M) \cos kM \; dM, \tag{684}$$

$$b_k := \frac{1}{\pi} \int_0^{2\pi} g(M) \sin kM \; dM, \tag{685}$$

Bessel functions of the first kind can be defined

$$J_k(x) := \frac{1}{2\pi}\int_0^{2\pi} \cos(kt - x\sin t)\, dt \tag{686}$$

We also will use the following recurrence relations

$$J_k(x) = \frac{x}{2k}[J_{k-1}(x) + J_{k+1}(x)], \tag{687}$$

$$J_k'(x) = \frac{1}{2}[J_{k-1}(x) - J_{k+1}(x)] \tag{688}$$

and the following approximations

$$J_1(x) = \frac{1}{2}x - \frac{1}{16}x^3 + \frac{1}{384}x^5 + \mathcal{O}(x^7), \tag{689}$$

$$J_2(x) = \frac{1}{8}x^2 - \frac{1}{96}x^4 + \mathcal{O}(x^6), \tag{690}$$

$$J_3(x) = \frac{1}{48}x^3 - \frac{1}{768}x^5 + \mathcal{O}(x^7), \tag{691}$$

$$J_4(x) = \frac{1}{384}x^4 + \mathcal{O}(x^6), \tag{692}$$

$$J_5(x) = \frac{1}{3840}x^5 + \mathcal{O}(x^7). \tag{693}$$

A.1 Eccentric Anomaly and True Anomaly

A.1.1 E

Let's try to expand $g(M) := e\sin E(M)$ as a Fourier series w.r.t. M, where the relationship between the mean and eccentric anomalies follows from the Kepler Equation $M = E - e\sin E$. Then,

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} e\sin[E(M)]\, dM. \tag{694}$$

Change of integration variable from M to E does not change the limits from $dM = (1 - e\cos E)\, dE$, we obtain

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} e\sin E(1 - e\cos E)\, dE = \frac{e}{2\pi}\int_0^{2\pi} \sin E\, dE - \frac{e^2}{2\pi}\int_0^{2\pi} \sin E\cos E\, dE = 0. \tag{695}$$

Since $\sin E(M)$ is anti-symmetric and $\cos kM$ is symmetric, we also get

$$a_k = \frac{1}{\pi}\int_0^{2\pi} e\sin[E(M)]\cos kM\, dM = 0. \tag{696}$$

On the other hand,

$$b_k = \frac{1}{\pi}\int_0^{2\pi} e\sin[E(M)]\sin kM\, dM \tag{697}$$

$$= \frac{1}{\pi}\int_0^{2\pi} e\sin E\sin[k(E - e\sin E)](1 - e\cos E)\, dE \tag{698}$$

We apply integration by parts $\int u\,dv = uv - \int v\,du$, where $u = e\sin E$ and

$$dv = \sin[k(E - e\sin E)](1 - e\cos E)\,dE = -\left\{\frac{1}{k}\frac{d}{dE}\cos[k(E - e\sin E)]\right\}dE. \tag{699}$$

Thus, $v = -\frac{1}{k}\cos[k(E - e\sin E)]$

$$b_k = -\Big|_0^{2\pi}\frac{e}{k\pi}\sin E\cos[k(E - e\sin E)] + \frac{e}{k\pi}\int_0^{2\pi}\cos[k(E - e\sin E)\cos E\,dE. \tag{700}$$

First term obviously disappears and for the second term, we can apply the trigonometric formula $\cos x\cos y = \frac{1}{2}\cos(x+y) + \frac{1}{2}\cos(x-y)$ to obtain

$$b_k = \frac{e}{2k\pi}\int_0^{2\pi}\cos[(k+1)E - ke\sin E]\,dE + \frac{e}{2k\pi}\int_0^{2\pi}\cos[(k-1)E - ke\sin E]\,dE. \tag{701}$$

Application of (686) and (687), yields

$$J_n(ke) = \frac{1}{2\pi}\int_0^{2\pi}\cos(nE - ke\sin E)\,dE. \tag{702}$$

and

$$b_k = \frac{e}{k}\left[J_{k+1}(ke) + J_{k-1}(ke)\right] = \frac{2}{k}J_k(ke), \tag{703}$$

Therefore, we obtain the series expansion

$$E = M + e\sin E = M + \sum_{k=1}^{\infty}\frac{2J_k(ke)}{k}\sin kM. \tag{704}$$

Expanding the individual terms

$$2J_1(e)\sin M \approx \left(e - \frac{1}{8}e^3 + \frac{1}{192}e^5 + \mathcal{O}(e^7)\right)\sin M \tag{705}$$

$$J_2(2e)\sin 2M \approx \left(\frac{1}{2}e^2 - \frac{1}{6}e^4 + \mathcal{O}(e^6)\right)\sin 2M \tag{706}$$

$$\frac{2}{3}J_3(3e)\sin 3M \approx \left(\frac{3}{8}e^3 - \frac{27}{128}e^5 + \mathcal{O}(e^7)\right)\sin 3M \tag{707}$$

$$\frac{1}{2}J_4(4e)\sin 4M \approx \left(\frac{1}{3}e^4 + \mathcal{O}(e^6)\right)\sin 4M \tag{708}$$

Thus, we finally obtain the approximation

$$\begin{aligned} E &\approx M + e\sin M + \frac{1}{2}e^2\sin 2M + e^3\left(-\frac{1}{8}\sin M + \frac{3}{8}\sin 3M\right) \\ &+ e^4\left(-\frac{1}{6}\sin 2M + \frac{1}{3}\sin 4M\right) + \mathcal{O}(e^5). \end{aligned} \tag{709}$$

A.1.2 $\cos nE$

Let now $g(M) := \cos[nE(M)]$. Then, clearly

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} \cos nE\, dM \tag{710}$$

$$= \frac{1}{2\pi}\int_0^{2\pi} \cos nE(1 - e\cos E)dE \tag{711}$$

$$= \frac{1}{2\pi}\int_0^{2\pi} \cos nE\, dE - e\frac{1}{2\pi}\int_0^{2\pi} \cos nE\cos E\, dE = -\frac{e}{2\pi}\frac{2\pi}{2}\delta_{n1} = -\frac{e}{2}\delta_{n1}, \tag{712}$$

$$b_k = \frac{1}{\pi}\int_0^{2\pi} \sin kM\cos nE\, dM = 0 \tag{713}$$

and by taking into account that $\cos kM = (1/k)d(\sin kM)/dM$, we also obtain

$$a_k = \frac{1}{\pi}\int_0^{2\pi} \cos kM\cos nE\, dM \tag{714}$$

$$= \frac{1}{k\pi}\Big|_0^{2\pi} \cos nE\sin kM - \frac{1}{k\pi}\int_0^{2\pi} \frac{d\cos nE}{dM}\sin kM\, dM \tag{715}$$

We perform a change of variables from M to E by noting that

$$\sin kM = \sin[k(E - e\sin E)], \tag{716}$$

$$\frac{d(\cos nE)}{dM} = -n\sin nE\frac{dE}{dM}. \tag{717}$$

Thereafter, substitution $\cos(x - y) - \cos(x + y) = 2\sin x\sin y$, yields

$$a_k = \frac{n}{k\pi}\int_0^{2\pi} \sin nE\sin[k(E - e\sin E)]\, dE \tag{718}$$

$$= \frac{n}{2k\pi}\int_0^{2\pi} \cos[(k - n)E - ke\sin E)]\, dE - \frac{n}{2k\pi}\int_0^{2\pi} \cos[(k + n)E - ke\sin E)]\, dE \tag{719}$$

$$= \frac{n}{k}\left[J_{k-n}(ke) - J_{k+n}(ke)\right] \tag{720}$$

and finally

$$\cos nE = -\frac{e}{2}\delta_{n1} + \sum_{k=1}^{\infty}\frac{n}{k}[J_{k-n}(ke) - J_{k+n}(ke)]\cos kM. \tag{721}$$

In the important special case $n = 1$

$$\cos E = -\frac{e}{2} + \sum_{k=1}^{\infty}\frac{2}{k}\frac{dJ_k}{de}(ke)\cos kM. \tag{722}$$

A.1.3 $\sin nE$

Let $g(M) = \sin[nE(M)]$. Then,

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} \sin nE\, dM \tag{723}$$

$$= \frac{1}{2\pi}\int_0^{2\pi} \sin nE(1 - e\cos E)\, dE \tag{724}$$

$$= \frac{1}{2\pi}\int_0^{2\pi} \sin nE\, dE - \frac{e}{2\pi}\int_0^{2\pi} \sin nE\cos E\, dE = 0. \tag{725}$$

$$a_k = \frac{1}{\pi}\int_0^{2\pi} \sin nE\cos kM\, dM = 0. \tag{726}$$

Using $\sin kM = -(1/k)(d/dM)\cos kM$, we can compute

$$b_k = \frac{1}{\pi}\int_0^{2\pi} \sin nE\sin kM\, dM \tag{727}$$

$$= -\frac{1}{k\pi}\Big|_0^{2\pi} \sin nE\cos kM + \frac{1}{k\pi}\int_0^{2\pi} \frac{d(\sin nE)}{dM}\cos kM\, dM \tag{728}$$

$$= \frac{n}{k\pi}\int_0^{2\pi} \cos nE\frac{dE}{dM}\cos kM\, dM \tag{729}$$

$$= \frac{n}{k\pi}\int_0^{2\pi} \cos nE\cos k(E - e\sin E)\, dE. \tag{730}$$

Using $\cos x\cos y = \frac{1}{2}\cos(x+y) + \frac{1}{2}\cos(x-y)$, we obtain

$$b_k = \frac{n}{2k\pi}\int_0^{2\pi} \cos[(k+n)E - ke\sin E)]\, dE + \frac{n}{2k\pi}\int_0^{2\pi} \cos[(k-n)E - ke\sin E)]\, dE \tag{731}$$

$$= \frac{n}{k}\left[J_{k+n}(ke) - J_{k-n}(ke)\right] \tag{732}$$

Thus, we finally obtain

$$\sin nE = \sum_{k=1}^{\infty} \frac{n}{k}\left[J_{k+n}(ke) - J_{k-n}(ke)\right]\sin kM \tag{733}$$

and in the special case $n = 1$

$$\sin E = \sum_{k=1}^{\infty} \frac{2}{k}\frac{dJ_k}{de}(ke)\sin kM \tag{734}$$

A.1.4 $\cos f$

A.1.5 $\sin f$

A.2 r/a

References

[1] Fitzpatrick, An Introduction to Celestial Mechanics, Cambridge University Press, 2012.

[2] Murray, Dermott, Solar System Dynamics, Cambridge University Press, 1999